

On Boundaries of Evidence

— Boundary-State Calculus for Typed Observation, Admissible Transfer, and Falsifiable Persistence

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Abstract

Boundary-State Calculus (BSC) is developed here as a typed audit calculus for finite open systems. Its purpose is not to supply a new ontology, but to decide when a claim remains warranted after boundary observation, quotienting, instrument change, transport, coarse-graining, dynamics, reconstruction, composition, and perturbation. We retain the authoritative BSC system objects and make explicit their measurable repair. We add a typed morphism record containing state and observable transport, stochastic simulation, equation residual, naturality defect, directed statistical deficiency, physical completion, and a certificate. Composition theorems show how defects and deficiencies accumulate and when a persistence claim must be demoted.

The calculus incorporates boundary trace and Dirichlet-to-Neumann response, operational quantum quotients, Blackwell–Le Cam comparison of experiments, Koopman residuals and ordered learnability limits, sheaf descent, structured cospan composition, non-invertible defects, and rigorous scale limits. It separates topological charge from physical charge by requiring an explicit bridge derived from a gauge structure, action, current, anomaly, boundary condition, or measurement protocol. A five-axis claim ledger and a proof-dependency graph prevent local success from promoting unrelated claims. Eight exact fixtures expose both admissible transfers and hard obstructions, including a deterministic regression receipt. The resulting formalism treats persistence as an operational organizing principle: a persistent object is a recursively maintained finite boundary-state, and persistent identity is operational equivalence maintained under repeated admissible folds across declared scales. This principle is defined and made falsifiable; it is not asserted as a universal law.

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1 Scope and non-claims

BSC concerns a finite open system only after “finite,” “open,” and “system” have been operationally fixed. Finiteness may mean bounded spatial domain, finite observation horizon, finite resources, or finite state cardinality; these meanings are not interchangeable. Openness means that a declared interface permits exchange with an exterior, not that all exterior degrees of freedom are known.

Definition 1.1 (Audit calculus). BSC is the collection of typed objects, admissibility conditions, residuals, loss measures, dependency rules, and certificates used to assess a claim about a finite open system under observation, representation, transfer, scale, dynamics, reconstruction, composition, and perturbation.

This definition is deliberately narrower than several interpretations that have accumulated around boundary language. BSC is not a fundamental ontology, a unified field theory, a proof of holography or quantum gravity, a theory of consciousness, or a license to identify systems by visual resemblance. A boundary may be geometric, causal, administrative, statistical, or instrumental; the type must be stated. The words “topological,” “recursive,” and “quantum” carry no evidential force without their mathematical objects and physical bridges.

Convention 1.2 (Local promotion). Evidence promotes only the claim node tested and those descendants whose declared dependencies are satisfied. It does not promote the framework, a neighboring claim, or an application by association.

Convention 1.3 (Source discipline). An internal derivation, peer-reviewed result, preprint, numerical execution, and analogy are distinct source states. A citation supplies provenance; it does not supply a missing map or proof.

The cited core corpus contains both durable mathematics and explicit later repairs. The relevant lineage is recorded in the source set [Boundary-State Research Program, 2026, Boundary-State Calculus, a,b, Tree, 2026a]. This paper keeps the durable core, records the repairs, and excludes civic or metaphysical material from mathematical inference. No empirical result is reported from the corpus.

2 Nomenclature, types, and front ledger

2.1 Basic types

Unless stated otherwise, measurable spaces are standard Borel. A Markov kernel $K : X \rightsquigarrow Y$ assigns a probability measure $K(x, \cdot)$ on Y measurably in x . Kernel composition is

$$(L \odot K)(x, A) = \int_Y L(y, A) K(x, dy).$$

A measurable map $f : X \rightarrow Y$ is identified with its Dirac kernel δ_f . We write $B_b(X)$ for the bounded Borel measurable real- or complex-valued functions on X . Norms, metrics, and units are declared locally; quantities of unlike units are never added merely by choosing coefficients.

Table 1: Core nomenclature and types.

Symbol	Type	Role
Ω_ℓ	bounded domain or finite carrier	physical or abstract interior at scale ℓ ; its meaning and units are declared
$\partial_\varepsilon \Omega_\ell$	measurable subdomain, or $\partial \Omega_\ell$ with trace regularity	finite sensor layer; never silently identified with an ideal boundary trace
$X_\ell, U_\ell, Y_\ell, C_\ell$	standard Borel spaces	raw states, controls, outputs, and contexts/calibrations
$\sim_\ell$	measurable equivalence relation on X_ℓ	observational, gauge, or representational equivalence; quotient $\bar{X}_\ell = X_\ell / \sim_\ell$ used only when standard Borel
$\mathcal{F}_{\ell,t}$	increasing sigma-algebras	legal information available immediately before time t 's outcome
$K_{\ell,t}^X, H_{\ell,t}^X$	$X_\ell \times U_\ell \times C_\ell \rightsquigarrow X_\ell, Y_\ell$	raw dynamics and observation kernels before quotient descent
$K_{\ell,t}^{\text{dyn}}$	$\bar{X}_\ell \times U_\ell \times C_\ell \rightsquigarrow \bar{X}_\ell$	descended local dynamics after lumpability; superscript prevents collision with interscale post-processing
$H_{\ell,t}^{\text{obs}}$	$\bar{X}_\ell \times U_\ell \times C_\ell \rightsquigarrow Y_\ell$	descended instrument channel after quotient invariance
G_ℓ	measurable loss/metric structure	comparison geometry with units and domain stated
$V_\ell \subseteq \bar{X}_\ell$	measurable admissible region	viability, safety, or model-validity domain
R_ℓ	declared family of typed kernels/maps	inherited class of scale, quotient, representation, or reconstruction maps; each use is namespaced below
B_ℓ	typed ledger	boundary flux, entropy, information, calibration, degeneracy, and provenance entries
$\mathcal{O}_\ell$	vector space, C^* -algebra, or measurable function class	declared accessible observables
$\tau_\partial, \Lambda_\Omega, E_\partial, e_\partial$	trace, response, reconstruction relation, selected extension	distinct boundary operators; their domains are not interchangeable
$\mathfrak{M}_{\ell m}$	typed morphism record	state/observable transfer, loss, defects, completion, and certificate
$\delta(E, F)$	number in $[0, 1]$	directed deficiency: ability of experiment E to simulate F
Cert	finite structured record	assumptions, tolerances, hashes, proof/execution receipts, sources, obligations, and status vector

2.2 Independent status axes

Definition 2.1 (Status vector). Every claim c carries

$$\text{Stat}(c) = (s_{\text{math}}, s_{\text{emp}}, s_{\text{comp}}, s_{\text{src}}, s_{\text{tr}}),$$

with the following partially ordered axes:

$$\begin{aligned} s_{\text{math}} &\in \{\text{ill-posed, refuted, conjectural, conditional, proved}\}, \\ s_{\text{emp}} &\in \{\text{contradicted, untested, single study, replicated}\}, \\ s_{\text{comp}} &\in \{\text{failed, unexecuted, executed, exact receipt}\}, \\ s_{\text{src}} &\in \{\text{unchecked, internal, present proof, verified preprint, verified publication}\}, \\ s_{\text{tr}} &\in \{\text{blocked, local only, bounded, certified}\}. \end{aligned}$$

Within each displayed set, entries are ordered from least to greatest support; the resulting finite chains supply the meet used below. The ordering is for evidential propagation: it does not identify “ill-posed” with “false” or “contradicted” with “untested.” An axis may instead be marked not applicable, in which case no edge may depend on it. Coordinates are never averaged.

The ordering is evidential, not rhetorical. For example, a proved theorem can remain empirically untested; an exact execution can implement an ill-posed physical encoding; a peer-reviewed source can be irrelevant to the asserted bridge.

Table 2: Concise claim-status ledger at entry. Every cell uses exactly one status from [definition 2.1](#), or N/A when that axis is inapplicable.

Claim	Math.	Empirical	Computation	Source	Transfer
Kernel quotient descent	proved	N/A	N/A	present proof	certified
Morphism defect accumulation	proved	N/A	unexecuted	present proof	bounded
Universal persistence principle	conjectural	untested	unexecuted	internal	local only
Universal boundary reconstruction	refuted	N/A	unexecuted	present proof	blocked
Specified Koopman impossibility classes	proved	N/A	N/A	verified publication	local only
Winding determines electric charge	refuted	untested	unexecuted	present proof	blocked
Klingman toroidal model yields QCD	ill-posed	untested	unexecuted	verified publication	blocked
Fixtures F1–F7	proved	N/A	unexecuted	present proof	local only
Fixture F8 universal claim	refuted	N/A	exact receipt	present proof	local only

2.3 Dependency graph

A claim ledger is a directed acyclic graph $D = (\mathcal{C}, E)$. An edge $a \xrightarrow{J} b$ states that claim b depends on claim a on the axis set $J \subseteq \{\text{math, emp, comp, src, tr}\}$. Status propagates only along named axes.

Definition 2.2 (Demotion operator). For a claim b , let $\text{Req}_j(b)$ be its immediate predecessors on axis j . Its admissible status is bounded above by

$$s_j(b) \preceq \bigwedge_{a \in \text{Req}_j(b)} s_j(a)$$

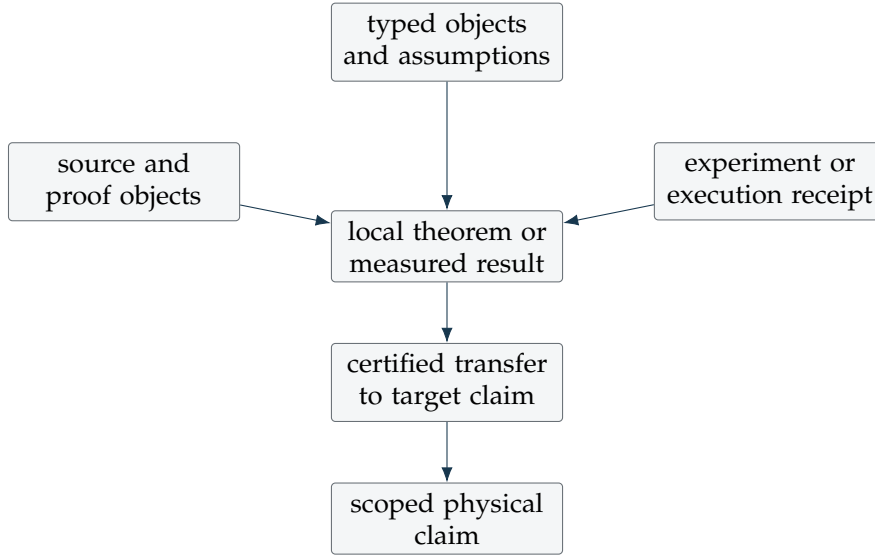


Figure 1: Minimal dependency architecture. Failure of an upstream requirement demotes only descendants reached by the edge’s declared status axes.

unless b contains an independent certificate that discharges the dependency. New contradictory evidence applies the meet again. Missing type data makes the mathematical status ill-posed; missing execution makes only the computational status unexecuted; a failed transfer makes only the deployment status blocked unless it also supplies a mathematical counterexample.

3 Finite boundary-state systems

3.1 Retained authoritative objects

The original system object is retained exactly:

$$S_\ell(t) = (\Omega_\ell, \partial_\varepsilon \Omega_\ell, X_\ell(t) / \sim_\ell, U_\ell(t), Y_\ell(t), H_\ell, G_\ell, V_\ell, R_\ell, K_\ell, B_\ell, \text{rent}_\ell). \quad (1)$$

It records a finite domain, observed boundary layer, quotient state, controls, outputs, observation, comparison geometry, viability, transformation class, local dynamics, boundary ledger, and an admission penalty [Boundary-State Research Program, 2026]. It is historically authoritative but under-typed: the context used by H_ℓ and the legal filtration are absent, R_ℓ is overloaded, and rent_ℓ has no fixed codomain.

The corpus then makes the following measurable repair, also retained exactly:

$$\hat{S}_\ell = (\Omega_\ell, \partial_\varepsilon \Omega_\ell, X_\ell, \sim_\ell, U_\ell, Y_\ell, C_\ell, \mathcal{F}_\ell, K_\ell, H_\ell, G_\ell, V_\ell, R_\ell, B_\ell). \quad (2)$$

Volume III explicitly works with (2) while retaining the application layers of the first object [Boundary-State Calculus, a,b]. We therefore take $\hat{S}_\ell$, not a newly invented tuple, as the current core.

Definition 3.1 (Typed interpretation of $\hat{S}_\ell$). Ω_ℓ is a declared bounded domain or finite carrier; $\partial_\varepsilon \Omega_\ell$ is a measurable layer, with ε carrying the domain’s length or resolution unit. $X_\ell, U_\ell, Y_\ell, C_\ell$ are standard Borel spaces. In particular, C_ℓ is a structured physical-configuration space containing the instrument, observer or reference-frame state, controller, clock, and calibration state needed by the experiment; omitting one of these factors is a declared quotient, not a claim that the factor is nonphysical. $\sim_\ell$ is a measurable exact equivalence and $\tilde{X}_\ell = X_\ell / \sim_\ell$ is required to

be standard Borel whenever ordinary quotient kernels are used. $\mathcal{F}_{\ell,t}$ is an increasing filtration.

$$K_{\ell,t}^X : X_\ell \times U_\ell \times C_\ell \rightsquigarrow X_\ell, \quad H_{\ell,t}^X : X_\ell \times U_\ell \times C_\ell \rightsquigarrow Y_\ell$$

are the raw experiment kernels. Write $q_\ell : X_\ell \rightarrow \bar{X}_\ell$. The observation kernel must be invariant,

$$x \sim_\ell x' \implies H_{\ell,t}^X(\cdot \mid x, u, c) = H_{\ell,t}^X(\cdot \mid x', u, c),$$

and the dynamics must be lumpable,

$$x \sim_\ell x' \implies (q_\ell)_\# K_{\ell,t}^X(\cdot \mid x, u, c) = (q_\ell)_\# K_{\ell,t}^X(\cdot \mid x', u, c).$$

Only then do quotient descent and the final sigma-algebra define the tuple's operational kernels

$$K_{\ell,t}^{\text{dyn}} : \bar{X}_\ell \times U_\ell \times C_\ell \rightsquigarrow \bar{X}_\ell, \quad H_{\ell,t}^{\text{obs}} : \bar{X}_\ell \times U_\ell \times C_\ell \rightsquigarrow Y_\ell.$$

G_ℓ is a declared metric, divergence, or measurable positive semidefinite bilinear-form field; $V_\ell \subseteq \bar{X}_\ell$ is measurable. R_ℓ is the inherited registry of typed scale, quotient, representation, and reconstruction maps, never a single polymorphic arrow. B_ℓ is a schema of ledger entries, each with a value, unit, method, uncertainty, time, and provenance. Legal preparation and feedback are kernels

$$\pi_{\ell,t} : (H_{\ell,t}, \mathcal{F}_{\ell,t}) \rightsquigarrow U_\ell \times C_\ell$$

from the measurable observation history $H_{\ell,t}$; their adaptedness means that $\pi_{\ell,t}$ depends only on information in $\mathcal{F}_{\ell,t}$. A state-Markov policy is separately denoted $\bar{\pi}_{\ell,t} : \bar{X}_\ell \rightsquigarrow U_\ell \times C_\ell$. Such a policy, or a sufficient history augmentation of $\bar{X}_\ell$, is required before the joint kernels below induce a state-marginal fold.

Definition 3.2 (System certificate companion). Without changing (2), associate

$$\mathcal{C}_\ell = (D_\ell^{\text{rec}}, \text{Prov}_\ell, \text{Cert}_\ell),$$

where D_ℓ^{rec} contains reconstruction relations and their ambiguity classes; Prov_ℓ contains source identifiers, calibration lineage, data and code hashes; and Cert_ℓ contains assumptions, boundary conditions, tolerances, proof or execution objects, unresolved obligations, and Stat. The pair $(\hat{S}_\ell, \mathcal{C}_\ell)$ is an instantiated certified system, not a replacement system tuple.

Definition 3.3 (Experiment bundle). The retained experiment object is

$$\mathcal{E} = (S, \mathcal{T}, \mathcal{I}, \mathcal{M}_0, \mathcal{O}, \mathcal{N}, \mathcal{L}, \mathcal{A}, \mathcal{D}).$$

Here S is (2); the repaired target is

$$\mathcal{T}_{\text{src}} = (Y, H, \tau, \mathcal{F}_t^{\text{legal}}),$$

where the source explicitly calls H the horizon but leaves τ undefined. We set $h := H$, choose a declared finite sampling set $J_{t,h} \subset [t, t+h]$, and introduce a measurable target functional

$$q : (Y^{J_{t,h}}, \mathcal{B}(Y)^{\otimes J_{t,h}}) \rightarrow Z_q.$$

We then use the explicit repair

$$\mathcal{T}^{\text{rep}} = (Y, h, q, \mathcal{F}_t^{\text{legal}}).$$

The observation channel remains in S and is not duplicated in the target. $\mathcal{I}$ is the intervention class; $\mathcal{M}_0$ a baseline; $\mathcal{O}$ candidate operators; $\mathcal{N}$ matched nulls; $\mathcal{L}$ a scoring rule; $\mathcal{A}$ an admission rule; and

$$\mathcal{D} : \text{Claims} \times \text{Evidence} \rightarrow \{\text{admit, sandbox, watch, demote, retire}\}.$$

The name “demotion map” is retained, although its codomain also includes admission.

3.2 Typed folds

The original fold $\Pi_{V_m}^{G_m} \circ R_{\ell m} \circ H_\ell \circ K_\ell$ mixes state dynamics, observation, inference, and state transport. Its architecture is useful but its displayed composite is not type-correct.

Definition 3.4 (Lifted observation fold). Put $Z_\ell^{\text{sc}} = \bar{X}_\ell \times U_\ell \times C_\ell$. Define

$$\begin{aligned}\tilde{K}_\ell((x, u, c), d(x', u', c')) &= K_\ell^{\text{dyn}}(dx' \mid x, u, c) \delta_u(du') \delta_c(dc'), \\ \tilde{H}_\ell((x, u, c), d(y, c')) &= H_\ell^{\text{obs}}(dy \mid x, u, c) \delta_c(dc').\end{aligned}$$

If $R_{\ell m}^Y : Y_\ell \times C_\ell \rightsquigarrow \bar{X}_m$ is an inference kernel and $\Pi_{V_m}^{G_m} : \bar{X}_m \rightsquigarrow V_m$ a declared measurable repair or projection kernel, then

$$F_{\ell m}^{\text{obs}} = \Pi_{V_m}^{G_m} \odot R_{\ell m}^Y \odot \tilde{H}_\ell \odot \tilde{K}_\ell : Z_\ell^{\text{sc}} \rightsquigarrow V_m$$

is a typed observation-to-state fold. A direct state coarse-graining $R_{\ell m}^X : \bar{X}_\ell \rightsquigarrow \bar{X}_m$ defines a separate route

$$F_{\ell m}^X = \Pi_{V_m}^{G_m} \odot R_{\ell m}^X \odot \delta_{\pi_X} \odot \tilde{K}_\ell, \quad \pi_X : Z_\ell^{\text{sc}} \rightarrow \bar{X}_\ell.$$

Conflating these routes hides whether information was discarded by the instrument, the inference, the quotient, or the scale map.

4 Observation events, filtration, and leakage control

Definition 4.1 (Observation atom). The retained observation atom is

$$a_t = (\ell, t, \Omega, \partial_\varepsilon \Omega, x_t, u_t, c_t, H_t, y_t, \Sigma_{y,t}, \mathcal{F}_t).$$

It is legal for prediction horizon $h > 0$ when $x_t \in \bar{X}_\ell$, $y_t \in Y_\ell$, and $H_t = H_{\ell,t}^{\text{obs}}$. When $Y_\ell \subseteq \mathbb{R}^d$, $\Sigma_{y,t} \in \mathbb{R}^{d \times d}$ is positive semidefinite and has declared output units squared; for a general output space it is replaced by a typed uncertainty object. Every statistic used to predict Y_{t+h} is $\mathcal{F}_t$ -measurable.

Definition 4.2 (Boundary event). For $r \in \partial\Omega$,

$$e_t = (r, t, J_k(r, t), X_k(r, t), \nu(r), \tau(r))$$

records a flux/current component J_k , boundary field X_k , outward normal ν , and orientation or time label τ . Their tensor types and units must be fixed by the governing model before any conservation checksum is formed.

Assumption 4.3 (Pre-outcome freeze). For an outcome at time t , the design, preprocessing graph, dictionary, candidate operators, null library, thresholds, and seed policy are $\mathcal{F}_{t-}$ -measurable. Holdout outcomes and post-outcome edits are not.

Definition 4.4 (Leakage defect). Let Z denote the complete fitted artifact and Y^{hold} the held-out outcome. A protocol has structural leakage if Z is not measurable with respect to the declared pre-outcome sigma-algebra. A probabilistic diagnostic is

$$\lambda_{\text{leak}} = I(Z; Y^{\text{hold}} \mid \mathcal{F}_{t-}),$$

when the conditional mutual information exists. The diagnostic being zero does not prove correct measurability; positive value refutes isolation.

The earlier corpus condition that two sigma-algebras have empty intersection is impossible because both contain the empty set and sample space. The repair is conditional isolation plus a logged access history. If [assumption 4.3](#) fails, computational or empirical promotion based on that holdout is demoted regardless of predictive score.

Proposition 4.5 (Predictable sequential admission). *Let A_t be false-admission events and let α_t be $\mathcal{F}_{t-1}$ -measurable. If $P(A_t \mid \mathcal{F}_{t-1}) \leq \alpha_t$ almost surely and $\sum_t \alpha_t \leq \alpha$, then $P(\bigcup_t A_t) \leq \alpha$.*

Proof. Taking expectations gives $P(A_t) \leq \mathbb{E}\alpha_t$. The union bound and Tonelli's theorem give $P(\bigcup_t A_t) \leq \sum_t \mathbb{E}\alpha_t \leq \mathbb{E} \sum_t \alpha_t \leq \alpha$. $\square$

This result does not validate the null model, exchangeability, or the physical meaning of the score. Those are separate dependencies.

5 Boundary trace, response, and reconstruction

Let $\Omega \subset \mathbb{R}^n$ be bounded with boundary regularity stated locally. The following three operations are distinct.

Definition 5.1 (Trace and layer). For a Banach function space $\mathcal{X}(\Omega)$ admitting a continuous trace,

$$\tau_{\partial} : \mathcal{X}(\Omega) \rightarrow \mathcal{X}(\partial\Omega).$$

For finite sensor resolution,

$$\partial_{\varepsilon}\Omega = \{x \in \Omega : \text{dist}(x, \partial\Omega) \leq \varepsilon\}, \quad L_{\varepsilon} : \mathcal{X}(\Omega) \rightarrow \mathcal{Y}_{\varepsilon}$$

is a separate measurement map. No equality $L_{\varepsilon} = \tau_{\partial}$ is presumed.

Definition 5.2 (Boundary response). Let $L_a u = 0$ be a declared elliptic boundary-value problem, with coefficient a in an admissible class. Let $\mathcal{B}_D$ and $\mathcal{B}_N$ be the declared Dirichlet and conormal-response spaces. In the standard weak uniformly elliptic setting on a bounded Lipschitz domain one may take $\mathcal{B}_D = H^{1/2}(\partial\Omega)$ and $\mathcal{B}_N = H^{-1/2}(\partial\Omega)$. If the weak solution u_f is well posed for every $f \in \mathcal{B}_D$, the Dirichlet-to-Neumann map is

$$\Lambda_{a,\Omega} : \mathcal{B}_D \rightarrow \mathcal{B}_N, \quad f \mapsto \partial_{\nu,a} u_f|_{\partial\Omega},$$

where $\partial_{\nu,a}$ is the model's conormal derivative. Thus $\Lambda_{a,\Omega}$ maps imposed boundary data to measured response; it is not the trace of an arbitrary interior state.

Definition 5.3 (Reconstruction relation). For a data class $\mathcal{D}$ and admissible interior class $\mathcal{A}$,

$$E_{\partial} \subseteq \mathcal{D} \times (\mathcal{A}/\approx)$$

is a possibly set-valued reconstruction relation. The equivalence $\approx$ contains every gauge or boundary-fixing diffeomorphism ambiguity left invariant by the data. Identifiability means that each data item in the range has at most one equivalence class; stability is a separate bound

$$d_{\mathcal{A}/\approx}(a_1, a_2) \leq \omega(d_{\mathcal{D}}(\Lambda_{a_1}, \Lambda_{a_2}))$$

on a stated subset, for a declared modulus ω .

Definition 5.4 (Selected extension). For a declared subset $\mathcal{D}_0 \subseteq \mathcal{D}$, a selected extension is a measurable map

$$e_{\partial} : \mathcal{D}_0 \rightarrow \mathcal{X}(\Omega)$$

whose graph, after passage to the ambiguity class, lies in E_{∂} . Existence and measurability are assumptions; linearity and boundedness are additional assumptions, never consequences of the relation. Any quantity evaluated on $e_{\partial}(d)$ must either be invariant under $\approx$ or record the representative-selection dependence.

Scoped isotropic uniqueness theorems are available in the dimensions and regularity classes of the cited works [Sylvester and Uhlmann, 1987, Astala and Päiväranta, 2006]. In the anisotropic manifold class of Lassas and Uhlmann [2001], recovery is modulo the specified boundary-fixing geometric equivalence. Stability is weaker and separately conditional [Alessandrini, 1988]. None of these results licenses the phrase “the boundary determines the interior” without the PDE, data class, regularity, quotient, boundary conditions, and stability theorem.

Proposition 5.5 (Trace-limited prediction). *Let $F : \mathcal{X}(\Omega) \rightarrow Z$ be L_F -Lipschitz and let a selected extension $e_\partial L_\varepsilon x$ be defined. Then*

$$\|F(x) - F(e_\partial L_\varepsilon x)\|_Z \leq L_F \|x - e_\partial L_\varepsilon x\|_{\mathcal{X}}.$$

Proof. This is the defining Lipschitz inequality. It is an upper bound, not an identifiability theorem. $\square$

If additive data noise η is passed through a bounded linear selected extension, $\|e_\partial \eta\| \leq \|e_\partial\| \|\eta\|$. The operator norm supplies an upper worst-case amplification, not a universal lower error bound.

6 Observable algebras and operational quotients

6.1 Classical experiment quotients

Definition 6.1 (Exact observation equivalence). For legal contexts and controls,

$$x \sim_H x' \iff H^X(\cdot \mid x, u, c) = H^X(\cdot \mid x', u, c) \text{ for all legal } (u, c).$$

A query $q : X \rightarrow Z$ descends when it is constant on these classes.

Theorem 6.2 (Kernel quotient descent). *Let $q : X \rightarrow \bar{X} = X/\sim$ be a measurable quotient of standard Borel spaces, with the final sigma-algebra, and let $M : X \rightsquigarrow Y$ be a kernel constant on equivalence classes such that every coordinate $M(\cdot, A)$ is q -measurable. Then there is a unique kernel $\bar{M} : \bar{X} \rightsquigarrow Y$ satisfying $M = \bar{M} \odot \delta_q$.*

Proof. For A in a countable generator of $\mathcal{B}(Y)$, define $\bar{M}([x], A) = M(x, A)$. Constancy makes this well defined. Quotient measurability and a monotone-class argument extend it to every Borel A ; countable additivity is inherited pointwise. Surjectivity of q gives uniqueness. $\square$

If $\bar{X}$ is not standard Borel or stabilizer information matters, an ordinary quotient is inadequate; a measurable groupoid or invariant-algebra description is required.

Definition 6.3 (Confusability pseudometric). Given a probability metric d_P ,

$$d_E(x, x') = \sup_{(u, c) \text{ legal}} d_P\left(H^{\text{obs}}(\cdot \mid x, u, c), H^{\text{obs}}(\cdot \mid x', u, c)\right).$$

The set $d_E(x, x') \leq \varepsilon$ is an identified neighborhood or confusability graph relation. It is not generally transitive and therefore does not generally define a quotient.

6.2 Quantum operational quotients

Let $\mathcal{H}$ be finite-dimensional for the present definition, and let $\mathcal{A}_{\text{acc}} \subseteq \mathcal{B}(\mathcal{H})$ be the declared operator system of accessible effects.

Definition 6.4 (Operational quantum equivalence). For density operators ρ, σ ,

$$\rho \sim_{\mathcal{A}_{\text{acc}}} \sigma \iff \sup_{\substack{0 \leq E \leq I \\ E \in \mathcal{A}_{\text{acc}}}} |\text{Tr } E(\rho - \sigma)| = 0.$$

This is exact operational equivalence: equality of accessible probabilities, not equality of matrix representations. The pseudometric

$$d_{\mathcal{A}_{\text{acc}}}(\rho, \sigma) = \sup_{\substack{0 \leq E \leq I \\ E \in \mathcal{A}_{\text{acc}}}} |\text{Tr } E(\rho - \sigma)|$$

defines ε -confusability by $d_{\mathcal{A}_{\text{acc}}}(\rho, \sigma) \leq \varepsilon$. For $\varepsilon > 0$ confusability is generally nontransitive and is not called an equivalence.

A state channel $\Phi : \mathcal{T}(\mathcal{H}_\ell) \rightarrow \mathcal{T}(\mathcal{H}_m)$ travels with its Heisenberg pullback

$$\Phi^\sharp : \mathcal{B}(\mathcal{H}_m) \rightarrow \mathcal{B}(\mathcal{H}_\ell), \quad \text{Tr}[\Phi(\rho)A] = \text{Tr}[\rho \Phi^\sharp(A)].$$

Quantum reference-frame changes may alter which subalgebras count as subsystems or accessible observables [Carette et al., 2025, Ahmad et al., 2022, Castro-Ruiz and Oreshkov, 2025]. Discarding is therefore relative to a declared algebra and factorization, not an observer-free deletion operation.

Definition 6.5 (Transform–discard defect). Let $\Phi : \mathcal{T}(\mathcal{H}_R \otimes \mathcal{H}_S) \rightarrow \mathcal{T}(\mathcal{H}_{R'} \otimes \mathcal{H}_{S'})$ be a frame or description channel. Suppose a candidate reduced channel $\bar{\Phi} : \mathcal{T}(\mathcal{H}_S) \rightarrow \mathcal{T}(\mathcal{H}_{S'})$ is declared. On accessible effects $\mathcal{A}_{S'}$,

$$\eta_{\text{td}}(\rho) = \sup_{\substack{0 \leq A \leq I \\ A \in \mathcal{A}_{S'}}} |\text{Tr } A \text{ Tr}_{R'} \Phi(\rho) - \text{Tr } A \bar{\Phi}(\text{Tr}_R \rho)|.$$

Transform and discard commute operationally on a class $\mathcal{Q}$ when a single $\bar{\Phi}$ makes

$$\sup_{\rho \in \mathcal{Q}} \eta_{\text{td}}(\rho) = 0.$$

Proposition 6.6 (Paired transport preserves probabilities). *For every state ρ and target effect A , $\text{Tr}[\Phi(\rho)A] = \text{Tr}[\rho \Phi^\sharp(A)]$. Consequently a representation change is operationally harmless only for probabilities tested with the paired observable transport.*

Proof. This is the definition of the Banach adjoint $\Phi^\sharp$. □

7 Admissible morphisms and composition

The inherited deterministic morphism

$$\mu_{a \rightarrow b} = (R_{a \rightarrow b}, R_Y, \varepsilon_R, \tau_R, q_{a \rightarrow b})$$

paired state and output maps with an error field, an otherwise untyped τ_R , and quotient compatibility [Boundary-State Research Program, 2026]. In particular, τ_R is not silently promoted here to a certificate. The printed two-step error rule in that source is repaired to the type-correct downstream Lipschitz bound

$$\varepsilon_{ac} \leq \varepsilon_{bc} + L_{bc} \varepsilon_{ab}.$$

The later budgeted record is retained exactly as

$$\mu = (R_X, R_Y, R_U, R_C, \alpha_H, \alpha_K, q_{ab}, b_\mu),$$

where q_{ab} records quotient compatibility and b_μ a distortion budget [Boundary-State Calculus, a]. Under explicit typing, R_X embeds in $T_{\ell m}$, R_Y in $K_{\ell m}$, R_U, R_C in the completion, $\alpha_H, \alpha_K, b_\mu$ in the defect/tolerance fields, and q_{ab} in the information-loss and admissibility fields. The older $R_{a \rightarrow b}, R_Y$ embed similarly; $\varepsilon_R, \tau_R, q_{a \rightarrow b}$ are preserved in the certificate until their types are discharged. The following record extends both; it does not replace the system object.

Definition 7.1 (Typed BSC morphism). A morphism from certified system ℓ to certified system m is

$$\mathfrak{M}_{\ell \rightarrow m} = \left(T_{\ell m}, T_{\ell m}^\sharp, K_{\ell m}, R_{\ell m}, \Theta_{\ell m}, \delta_{\ell m}, C_{\ell m}, \text{Cert}_{\ell m} \right), \quad (3)$$

with a variant tag $v \in \{\text{det}, \text{stoch}, \text{quant}, \text{corr}, \text{defect}\}$. For the stochastic variant:

1. $T_{\ell m} : D_{\ell m} \rightsquigarrow \bar{X}_m$ is a Markov kernel on a declared measurable domain $D_{\ell m} \subseteq \bar{X}_\ell$.
- 2.

$$T_{\ell m}^\sharp : B_b(\bar{X}_m) \rightarrow B_b(D_{\ell m}), \quad T_{\ell m}^\sharp g(x) = \int g(x') T_{\ell m}(x, dx')$$

is positive and unital. It need not be multiplicative. In this bounded stochastic variant, $\mathcal{O}_i \subseteq B_b(\bar{X}_i)$, and the declared observable families must satisfy

$$T_{\ell m}^\sharp(\mathcal{O}_m) \subseteq \mathcal{O}_\ell|_{D_{\ell m}};$$

otherwise the state transfer does not transport the observables claimed. Unbounded observables require a separately declared invariant integrability domain.

3. $K_{\ell m} : Y_\ell \rightsquigarrow Y_m$ is an observation post-processing or simulation channel.
4. Given equation operators $\mathcal{E}_i : \text{Dom}(\mathcal{E}_i) \rightarrow Z_i$ and an equation-space map $S_{\ell m} : Z_\ell \rightarrow Z_m$, the residual record $R_{\ell m}$ contains the target-equation residual, its Banach space, norm, units, evaluation class, and bound. In the deterministic variant,

$$R_{\ell m}(x) = \mathcal{E}_m(T_{\ell m}x) - S_{\ell m}\mathcal{E}_\ell(x) \in Z_m.$$

For stochastic or nonlinear equations, the certificate must specify the law-level weak residual; no canonical expectation is presumed.

5. Put $Z_i^{\text{sc}} = \bar{X}_i \times U_i \times C_i$. The completion $C_{\ell m}$ contains control and context transports and a joint lift

$$\hat{T}_{\ell m} : \hat{D}_{\ell m} \rightsquigarrow Z_m^{\text{sc}}, \quad \hat{D}_{\ell m} \subseteq Z_\ell^{\text{sc}},$$

whose state marginal agrees with $T_{\ell m}$ under the declared source policy. It also contains the physical carrier, controller, observer, reference frame, instrument, clock map, resource interface, and boundary conditions.

6. On the completed common domain,

$$\Theta_{\ell m} = \sup_{z \in \hat{D}_{\ell m}} d_m \left(H_m^{\text{obs}} \odot \hat{T}_{\ell m}(z), K_{\ell m} \odot H_\ell^{\text{obs}}(z) \right)$$

is a naturality or commuting-square defect in a declared probability metric.

7. For experiments $E_i = \{P_{i,\theta} : \theta \in \Theta\}$,

$$\delta_{\ell m} = \delta(E_\ell, E_m) = \inf_{K: Y_\ell \rightsquigarrow Y_m} \sup_{\theta \in \Theta} \|KP_{\ell,\theta} - P_{m,\theta}\|_{\text{TV}}.$$

The total-variation convention is $\|P - Q\|_{\text{TV}} = \frac{1}{2}\|p - q\|_1$ in the finite case. For the particular implemented post-processor in the record, define

$$e_{\ell m}(K_{\ell m}) = \sup_{\theta \in \Theta} \|K_{\ell m}P_{\ell,\theta} - P_{m,\theta}\|_{\text{TV}} \geq \delta_{\ell m}.$$

The certificate says whether this channel attains the infimum or is only a declared near-minimizer.

8. $\text{Cert}_{\ell m}$ is the record specified in [definition 3.2](#), augmented by variant, domains, tolerances, information destroyed, residual and defect evaluations, implementation evidence, and status vector.

A partial deterministic map uses the same record with explicit domain and failure output. A correspondence $T \subseteq X_\ell \times X_m$ requires a selection rule, multiplicity semantics, or probability kernel before observable pullback is single-valued. A quantum channel uses trace-class states and its completely positive unital adjoint. Defect and fusion variants are treated in [section 12](#). These variants do not share a single unqualified composition theorem.

Definition 7.2 (Admissibility). $\mathfrak{M}_{\ell m}$ is admissible for a declared claim only if its certificate answers all of the following:

1. the domain, codomain, variant, regularity, units, and boundary conditions of $T_{\ell m}$;
2. the target observable class and reverse-direction transport $T_{\ell m}^\sharp$;
3. the equivalence classes or information distinctions destroyed;
4. the target equations, equation-space map, residual norm, and tolerance;
5. the observation and other diagrams tested, including nonzero defects;
6. the physical carrier, controller, clock, observer, reference frame, and instrument that implement the transfer;
7. the assumptions, sources, proof objects, execution receipts, hashes, unresolved obligations, and five-axis status.

Failure to supply an item is a blocked transfer, not a small numerical error.

7.1 Composition in the stochastic variant

Assumption 7.3 (Compatible completions). For $\mathfrak{M}_{\ell m}$ and $\mathfrak{M}_{mn}$, the target interface, clock, resource contract, boundary conditions, control/context law, and carrier output of $C_{\ell m}$ match the source requirements of C_{mn} . The composite completion is their specified fiber product over the shared interface. Moreover, $\hat{T}_{\ell m}(z, \hat{D}_{mn}) = 1$ for every $z \in \hat{D}_{\ell m}$ in the composite domain. Correlated failure modes are retained.

Definition 7.4 (Stochastic composite). Under [assumption 7.3](#), define

$$T_{\ell n} = T_{mn} \odot T_{\ell m}, \quad K_{\ell n} = K_{mn} \odot K_{\ell m},$$

on the maximal measurable domain

$$D_{\ell n} = \{x \in D_{\ell m} : T_{\ell m}(x, D_{mn}) = 1\}.$$

Then

$$T_{\ell n}^\sharp = T_{\ell m}^\sharp \circ T_{mn}^\sharp.$$

On state–control–context space the compatible completion defines

$$\hat{T}_{\ell n} = \hat{T}_{mn} \odot \hat{T}_{\ell m}.$$

The certificate is the conjunction of assumptions and obligations, the union of provenance DAGs, and the axiswise meet of statuses. A new hash identifies the composite record; it is not a new proof.

Theorem 7.5 (Observation-defect propagation). Let $d_i = \|\cdot\|_{\text{TV}}$, and let $\eta(K_{mn}) \leq 1$ be the Dobrushin contraction coefficient of K_{mn} . For compatible stochastic morphisms,

$$\Theta_{\ell n} \leq \Theta_{mn} + \eta(K_{mn})\Theta_{\ell m}.$$

For a Wasserstein metric, the same statement holds with the declared Lipschitz coefficient of K_{mn} , provided the required moments exist.

Proof. Apply the triangle inequality through the intermediate law $K_{mn} \odot H_m^{\text{obs}} \odot \hat{T}_{\ell m}$. The first difference is bounded by Θ_{mn} after averaging over $\hat{T}_{\ell m}$; the second contracts by $\eta(K_{mn})$. The identity $\hat{T}_{\ell n} = \hat{T}_{mn} \odot \hat{T}_{\ell m}$ supplies the other side of the composite square. Take the supremum. $\square$

Theorem 7.6 (Directed deficiency triangle). *Suppose all three experiments have the same parameter set Θ , or have been pulled back to one by declared parameter maps. Then*

$$\delta(E_\ell, E_n) \leq \delta(E_\ell, E_m) + \delta(E_m, E_n).$$

Proof. Choose channels within ε of the two infima and compose them. Total variation contracts under Markov kernels; the triangle inequality gives the displayed bound plus 2ε . Let $\varepsilon \downarrow 0$. $\square$

Theorem 7.7 (Deterministic equation-residual propagation). *For deterministic state maps with $T_{\ell n} = T_{mn}T_{\ell m}$ and $S_{\ell n} = S_{mn}S_{\ell m}$, suppose $S_{mn} : Z_m \rightarrow Z_n$ is bounded linear, the equation domains are compatible, and $T_{\ell m}(D_{\ell m}) \subseteq \text{Dom}(R_{mn})$. Then*

$$R_{\ell n}(x) = R_{mn}(T_{\ell m}x) + S_{mn}R_{\ell m}(x).$$

Consequently,

$$\|R_{\ell n}\|_{\infty, D_{\ell n}} \leq \|R_{mn}\|_{\infty, T_{\ell m}(D_{\ell m})} + \|S_{mn}\| \|R_{\ell m}\|_{\infty, D_{\ell m}}.$$

Proof. Add and subtract $S_{mn}\mathcal{E}_m(T_{\ell m}x)$ in $\mathcal{E}_n(T_{mn}T_{\ell m}x) - S_{mn}S_{\ell m}\mathcal{E}_\ell(x)$. $\square$

The identity morphism has $T = \delta_{\text{id}}$, $T^\sharp = \text{id}$, $K = \delta_{\text{id}}$, zero residual and defects, zero deficiency, identity completion, and a certificate containing the relevant system identities. Kernel associativity, reverse pullback composition, and the two fiber-product unit isomorphisms give the unit laws. If completion compatibility fails, the composite is undefined rather than high-error.

8 Defects, residuals, deficiency, and information loss

Residual, naturality defect, statistical deficiency, quotient loss, and viability failure live in different spaces. They are recorded as a vector, not collapsed into a mechanism score.

Definition 8.1 (Loss vector). For a morphism, let $\mathcal{R}_{\ell m}$ be its typed residual-record space, let $\mathcal{Q}_{\ell m}^{\text{res}}$ be a declared residual test class, and let

$$\text{ev}_{\ell m}^{\text{res}} : \mathcal{R}_{\ell m} \times \mathcal{Q}_{\ell m}^{\text{res}} \rightarrow [0, \infty]$$

be a declared measurable scalar evaluator with recorded units. Let $\mathcal{Q}_{\ell m}^{\text{law}}$ be a declared class of admissible input probability laws. Put

$$\mathcal{L}_{\ell m} = (\mathfrak{r}_{\ell m}, \Theta_{\ell m}, \delta_{\ell m}, e_{\ell m}(K_{\ell m}), \lambda_{\text{quot}}, \alpha_{\text{exit}}),$$

where

$$\mathfrak{r}_{\ell m} = \sup_{\zeta \in \mathcal{Q}_{\ell m}^{\text{res}}} \text{ev}_{\ell m}^{\text{res}}(R_{\ell m}, \zeta), \quad \lambda_{\text{quot}}(q) = \sup_{x \sim_\ell x'} d_q(q(x), q(x'))$$

for a measurable target query $q : X_\ell \rightarrow (Q_q, d_q)$, and $\alpha_{\text{exit}} = \sup_{\mu \in \mathcal{Q}_{\ell m}^{\text{law}}} P_\mu(\tau_{V_m} \leq h)$. Here τ_{V_m} is the first exit time from V_m under the declared target dynamics and adapted policy. An important deterministic specialization takes $\mathcal{Q}_{\ell m}^{\text{res}} = D_{\ell m}^{\text{eval}}$ and $\text{ev}_{\ell m}^{\text{res}}(R, x) = \|R(x)\|_{Z_m}$. A law-level stochastic residual instead requires its own weak-form evaluator; it is not inserted into the deterministic formula. Each coordinate has its own tolerance and propagation law.

Definition 8.2 (Blackwell order). Experiment E_ℓ Blackwell-dominates E_m , written $E_\ell \succeq_B E_m$, when a Markov kernel K satisfies $KP_{\ell,\theta} = P_{m,\theta}$ for every θ . Thus $\delta(E_\ell, E_m) = 0$: data from ℓ can simulate data from m . The converse deficiency may be positive.

This direction follows Blackwell [1953] and Le Cam [1964], while noting that conventions in the literature vary. Only when needed do we use

$$\Delta(E, F) = \max\{\delta(E, F), \delta(F, E)\}.$$

The directed deficiency itself is not a distance.

Proposition 8.3 (Decision-risk consequence). *For a bounded loss $0 \leq L \leq M$, if $\delta(E, F) \leq \varepsilon$, then, for every $\eta > 0$, every decision rule for F has a randomized rule based on E whose worst-case risk exceeds it by at most $M(\varepsilon + \eta)$ under the present total-variation convention. If the deficiency infimum is attained and equals ε , one may take $\eta = 0$.*

Proof. Use a channel within η of the deficiency infimum and then apply the target decision rule. The expectation of a function in $[0, M]$ changes by at most $M\|P - Q\|_{TV}$. $\square$

Definition 8.4 (Boundary sufficiency gates). For a target Y_{t+h} , boundary data $X_{\partial,t}$, observed interior $X_{\Omega,t}^{\text{obs}}$, exterior $X_{\text{ext},t}$, and legal covariates Z_t , define

$$G_\partial = I(Y_{t+h}; X_{\partial,t} \mid Z_t), \quad S_{\text{ext}} = I(Y_{t+h}; X_{\text{ext},t} \mid X_{\partial,t}, X_{\Omega,t}^{\text{obs}}, Z_t).$$

All logarithms are base 2, so $G_\partial, S_{\text{ext}}, g_0, s_0$ are in bits. A target-relative statistical boundary claim requires $G_\partial \geq g_0$ and $S_{\text{ext}} \leq s_0$ separately. A large gain cannot compensate for an open exterior channel.

These are predictive diagnostics, not Sobolev traces, response maps, causal carrier proofs, or coefficient reconstruction theorems.

9 Dynamics, recurrence, Koopman operators, and learnability

Let $F : X \rightarrow X$ be measurable on a probability space $(X, \mathcal{B}, \mu)$. When F preserves μ , the Koopman operator on $L^2(\mu)$ is the isometry

$$[\mathcal{K}_F g](x) = g(F(x)). \quad (4)$$

It acts on observables, not on states [Koopman, 1931].

Definition 9.1 (Certified approximate mode). For a declared observable space $\mathcal{G} \subseteq L^2(\mu)$, $\phi \in \mathcal{G}$ normalized by $\|\phi\|_{L^2(\mu)} = 1$, and $\lambda \in \mathbb{C}$,

$$\varepsilon_{\text{dyn}}(\phi, \lambda) = \|\mathcal{K}_F \phi - \lambda \phi\|_{L^2(\mu)}.$$

A nonnormalized nonzero mode is assessed by the corresponding relative residual, dividing by $\|\phi\|_{L^2(\mu)}$; $\phi = 0$ is inadmissible. A certificate also states the dictionary, data-access model, norm, sampling law, finite time horizon, limiting order, perturbation class, stability bound, and learnability or Solvability Complexity Index class.

Small training error is not this residual unless the sampling theorem and dictionary-to-operator bridge are proved. A finite compression can generate spectral pollution. Residual DMD supplies a posteriori rejection and convergent spectral procedures for specified information models [Colbrook and Townsend, 2024, Colbrook et al., 2023]. It does not certify arbitrary learned modes.

Definition 9.2 (Recurrence and persistence). Let (Q, d_Q) be Polish, let $p \geq 1$, and let $q_i : \bar{X}_i \rightarrow Q$ be declared identity observables. Assume all pushforward laws below lie in $\mathcal{P}_p(Q)$. For each i , a completed admissible fold $F_i : Z_i^{\text{sc}} = \bar{X}_i \times U_i \times C_i \rightsquigarrow \bar{X}_{i+1}$ and a certified filtration-adapted state-Markov policy $\bar{\pi}_i$ induce the state-marginal kernel

$$P_i(x, A) = \int_{U_i \times C_i} F_i((x, u, c), A) \bar{\pi}_i(d(u, c) \mid x).$$

History-dependent policies require history augmentation of $\bar{X}_i$; without that augmentation the displayed Markov kernel is not asserted. Let $\mathbf{X}_0 \sim \mu \in \mathcal{P}(\bar{X}_0)$ and $\mathbf{X}_{i+1} \mid \mathbf{X}_i \sim P_i(\mathbf{X}_i, \cdot)$. Denote the induced path law by P_μ , and define

$$\tau_V = \inf\{i \geq 0 : \mathbf{X}_i \notin V_i\}, \quad \inf \emptyset = \infty.$$

Endpoint recurrence of a law μ after n folds is

$$\text{Rec}_{q,n}(\mu) = W_{p,Q}((q_0)_\# \mu, (q_n)_\# (\mu P_0 \cdots P_{n-1})).$$

Persistent identity through n folds requires, for every prefix $k \leq n$,

$$W_{p,Q}((q_0)_\# \mu, (q_k)_\# (\mu P_0 \cdots P_{k-1})) \leq \varepsilon_k, \quad P_\mu(\tau_V \leq k) \leq \alpha_k^{\text{cum}},$$

where α_k^{cum} is a cumulative exit budget, and every fold's morphism certificate remains admissible. Persistence is therefore stronger than return at the final horizon.

This formalizes, as an organizing principle, the statement: *a persistent object is a recursively maintained finite boundary-state; persistent identity is an operational equivalence maintained under repeated admissible folds across declared scales*. It remains a target-relative definition and conjectural physical principle, not a universal law.

Theorem 9.3 (Prefix error budget). *Suppose the identity discrepancy satisfies $E_{i+1} \leq L_i E_i + e_i$, where $L_i \geq 0$ and $e_i \geq 0$ have common units. Then*

$$E_n \leq \left(\prod_{j=0}^{n-1} L_j \right) E_0 + \sum_{i=0}^{n-1} e_i \prod_{j=i+1}^{n-1} L_j.$$

Persistence is invalid at the first prefix whose bound exceeds its declared tolerance, even if later contraction brings the endpoint below tolerance.

Proof. Induction on n . The prefix condition follows from [definition 9.2](#). □

Directed deficiencies satisfy the separate additive bound of [theorem 7.6](#); viability failures satisfy

$$P(\tau_V \leq n) \leq \sum_{i=0}^n \beta_i$$

without independence whenever the per-step bounds $P(\mathbf{X}_i \notin V_i) \leq \beta_i$ hold. Alternatively the cumulative budget in [definition 9.2](#) is checked directly as $P(\tau_V \leq n) \leq \alpha_n^{\text{cum}}$. These coordinates must not be folded into E_n .

9.1 Learnability boundary

The Solvability Complexity Index records how many nested limiting processes a specified computational problem requires and whether error control is available [Colbrook and Hansen, 2023]. Recent Koopman results give matching success and impossibility regimes: for broad declared classes, some spectral tasks require ordered limits, and no universal single-limit learner can solve the stated problems; stronger structure restores certified algorithms [Colbrook et al., 2026]. These are task-class impossibility results, not impossibility of all forecasting.

Convention 9.4 (Koopman claim gate). A recurrence claim based on a learned operator is blocked unless it identifies: (i) discrete eigenvalue, continuous spectrum, pseudospectrum, or spectral measure as the target; (ii) the observable dictionary and its density claim; (iii) the full or sampled residual; (iv) perturbation stability; (v) horizon and sampling law; (vi) the order of limits; and (vii) the applicable positive or negative learnability theorem.

10 Scale transfer, coarse-graining, and ordered limits

Definition 10.1 (Scale-limit certificate). A micro-to-macro claim must state the record

(ε , limit order, scaling, topology, boundary conditions,
time interval, uniformity domain, effective parameters, error, excluded regimes).

If more than one parameter tends to a limit, the certificate records whether the limit is joint, iterated, or path-dependent.

Theorem 10.2 (Periodic elliptic homogenization). *Let $\Omega \subset \mathbb{R}^d$ be bounded and Lipschitz, $Y = (0, 1)^d$, and let $A : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ be measurable, symmetric, and Y -periodic, with*

$$\lambda |\xi|^2 \leq \xi^\top A(y) \xi \leq \Lambda |\xi|^2 \quad \text{a.e. } y, \quad \xi \in \mathbb{R}^d,$$

for fixed $0 < \lambda \leq \Lambda < \infty$. For fixed $f \in H^{-1}(\Omega)$, let $u_\varepsilon \in H_0^1(\Omega)$ be the unique weak solution of

$$-\nabla \cdot (A(x/\varepsilon) \nabla u_\varepsilon) = f.$$

For $j = 1, \dots, d$, let $\chi_j \in H_{\text{per}}^1(Y)/\mathbb{R}$ solve

$$\int_Y A(y)(e_j + \nabla \chi_j) \cdot \nabla v \, dy = 0 \quad \text{for all } v \in H_{\text{per}}^1(Y)/\mathbb{R},$$

and define

$$A_{\text{hom}} e_j = |Y|^{-1} \int_Y A(y)(e_j + \nabla \chi_j) \, dy.$$

As $\varepsilon \downarrow 0$, after A, Ω, f are fixed,

$$u_\varepsilon \rightharpoonup u_0 \text{ in } H_0^1(\Omega), \quad u_\varepsilon \rightarrow u_0 \text{ in } L^2(\Omega),$$

and

$$A(x/\varepsilon) \nabla u_\varepsilon \rightharpoonup A_{\text{hom}} \nabla u_0 \quad \text{in } L^2(\Omega; \mathbb{R}^d),$$

where $u_0 \in H_0^1(\Omega)$ solves $-\nabla \cdot (A_{\text{hom}} \nabla u_0) = f$.

Proof. Uniform ellipticity and Lax–Milgram give a uniform H_0^1 bound. Weak compactness and Rellich compactness supply subsequential weak H_0^1 and strong L^2 limits. Periodic corrector test functions in the weak formulation identify every subsequential flux limit with $A_{\text{hom}} \nabla u_0$; uniqueness of the effective weak solution removes subsequence dependence. This is the standard corrector argument under the displayed hypotheses [Bensoussan et al., 1978]. $\square$

This stationary theorem has no time limit. It supplies no quantitative rate under the stated hypotheses and no uniformity over changing coefficient classes, data, or domains. Degenerate, nonlinear, random, oscillating-boundary, and coupled time-dependent regimes are excluded. Stochastic homogenization similarly exposes stationarity, mixing, dimension, and ellipticity assumptions [Gloria and Otto, 2012, Armstrong and Smart, 2016]. This is the model for BSC scale claims: no universal rate is inherited from the word “coarse-graining.”

Definition 10.3 (Dynamical scale defect). Let $(\bar{X}_b, d_b)$ be Polish. For a measurable state map $R_{ab}^X : \bar{X}_a \rightarrow \bar{X}_b$, source and target Markov kernels P_a, P_b , and an input law μ such that both displayed laws lie in $\mathcal{P}_p(\bar{X}_b)$, define

$$T_{ab}^{\text{dyn}}(\mu) = W_p\left((R_{ab}^X)_\#(\mu P_a), ((R_{ab}^X)_\#\mu)P_b\right).$$

Proposition 10.4 (Iterated scale defect). Suppose P_a, P_b preserve the required p -moments,

$$T_{ab}^{\text{dyn}}(\mu P_a^j) \leq \tau \quad \text{for } j = 0, \dots, n-1,$$

and P_b is L_b -Lipschitz in the Wasserstein distance induced by d_b on every propagated target law. Then

$$W_p\left((R_{ab}^X)_\#(\mu P_a^n), ((R_{ab}^X)_\#\mu)P_b^n\right) \leq \tau \sum_{k=0}^{n-1} L_b^k.$$

Proof. Insert the one-step commuting square at each time and iterate the Wasserstein triangle and Lipschitz inequalities. $\square$

Classical kinetic-to-fluid limits exhibit the necessary scope [Bardos et al., 1993, Golse and Saint-Raymond, 2004]. The 2025 hard-sphere program of Deng et al. [2025] is a preprint result for a specified hard-sphere $\rightarrow$ Boltzmann $\rightarrow$ Euler/Navier–Stokes–Fourier chain on specified tori and ordered regimes, together with the long-time hard-sphere derivation in Deng et al. [2024]. It is not a derivation of all continuum mechanics or a settlement of every reading of Hilbert’s sixth problem. BSC uses it as a case in disciplined scope, not as a universal bridge.

11 Local-to-global consistency and sheaf obstruction

Let $\mathcal{U}$ be a base of contexts closed under the finite intersections and relevant unions used below, or replace it by the topology generated by the declared cover. More generally, the same definitions may be made on a site after its Grothendieck topology is specified.

Definition 11.1 (Observation sheaf). An observation sheaf $\mathcal{F}$ assigns to every context $U \in \mathcal{U}$ a set, vector space, or probability model $\mathcal{F}(U)$, and to $V \subseteq U$ a restriction $\rho_V^U : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$, with identity and composition laws. Local sections $s_i \in \mathcal{F}(U_i)$ are compatible when

$$\rho_{U_i \cap U_j}^{U_i} s_i = \rho_{U_i \cap U_j}^{U_j} s_j$$

for every nonempty overlap. The sheaf axiom states that compatible sections have a unique global section $s \in \mathcal{F}(\bigcup_i U_i)$ restricting to them.

The last statement is an axiom of a chosen sheaf, not a fact about every data assignment. In contextual models one instead asks whether a compatible family or local relation admits a global section; failure is the obstruction [Abramsky and Brandenburger, 2011].

Definition 11.2 (Consistency operators). For a finite cell complex with finite-dimensional inner-product stalks, let C^0 be vertex assignments, C^1 edge discrepancy spaces, and $d^0 : C^0 \rightarrow C^1$ the signed consistency difference constructed from the cellular sheaf's coface maps. This cellular coface convention is covariant along face incidence and must not be confused with the contravariant restriction maps of the ordinary sheaf in [definition 11.1](#). The sheaf Laplacian is

$$L_{\mathcal{F}} = (d^0)^* d^0.$$

Then $\ker L_{\mathcal{F}} = \ker d^0$ is the space of exactly consistent global assignments. For data $s \in C^0$, $\|d^0 s\|$ is a declared consistency residual. Its scale depends on the chosen inner products.

This construction follows the finite cellular hypotheses of [Hansen and Ghrist \[2019\]](#); it is not a canonical scalar for arbitrary heterogeneous data. For torsor-valued or parity constraints, a cocycle class in H^1 may obstruct gluing, but cohomology is invoked only after the coefficient system and differential have been constructed. The corpus's earlier "scale cohomology" lacked these objects and is here demoted to path or naturality defect.

Proposition 11.3 (Gluing gate). *For an actual sheaf $\mathcal{F}$, pairwise compatible local sections glue uniquely. For a presheaf, relation-valued assignment, or noisy family, local nonemptiness alone does not imply a global section.*

Proof. The first statement is the sheaf axiom. The second is witnessed by [worked fixture 16.6](#). $\square$

12 Open-system and non-invertible composition

12.1 A concrete interface category

Let $\mathcal{T}$ be a set of port types, including units and direction, and let $\mathbf{l}_{\mathcal{T}}$ be the full subcategory of $\mathbf{Set}/\mathcal{T}$ on maps $p : A \rightarrow \mathcal{T}$ with finite domain. Thus it is the category of finite typed interfaces even when $\mathcal{T}$ itself is infinite. Let $\mathbf{C}$ be a category of internal carriers with finite colimits, and let $L : \mathbf{l}_{\mathcal{T}} \rightarrow \mathbf{C}$ embed an interface as a free boundary carrier.

Definition 12.1 (BSC decoration datum). Fix one mode $r \in \{\text{det}, \text{stoch}, \text{nondet}\}$ and one compatible time object $\mathbb{T}$. A BSC decoration datum is a lax symmetric monoidal functor

$$\mathcal{D}_{r,\mathbb{T}} : (\mathbf{C}, +, 0) \longrightarrow (\mathbf{Set}, \times, 1)$$

with laxator $\varphi_{N,M} : \mathcal{D}(N) \times \mathcal{D}(M) \rightarrow \mathcal{D}(N + M)$ and unit $\varphi_0 : 1 \rightarrow \mathcal{D}(0)$. An element of $\mathcal{D}(N)$ is a record

$$d_N = (\widehat{S}_N, r, \mathbb{T}, \text{Beh}_N, \text{Cert}_N),$$

where Beh_N is respectively a map/flow, Markov kernel, or trajectory relation with typed boundary ports. The functorial action $\mathcal{D}(f)$ specifies how behavior and certificates are transported along a carrier map f . Illegal unit, clock, boundary, or resource identifications map to a distinguished absorbing failed decoration and are blocked by admissibility. Mixed modes or clocks require a separately declared lax monoidal comparison functor; they are not composed by default.

Definition 12.2 (Typed open BSC system). A deterministic, stochastic, or nondeterministic open system of the fixed mode and clock from A to B is an isomorphism class of structured cospans

$$L(A) \xrightarrow{i} N \xleftarrow{o} L(B)$$

decorated by $d_N \in \mathcal{D}_{r,\mathbb{T}}(N)$.

Given $A \rightarrow N \leftarrow B$ and $B \rightarrow M \leftarrow C$, form the pushout $P = N +_{L(B)} M$ in $\mathbf{C}$, with coprojections $j_N : N \rightarrow P$, $j_M : M \rightarrow P$. The composite is

$$L(A) \longrightarrow N +_{L(B)} M \longleftarrow L(C),$$

with decoration

$$d_P = \mathcal{D}([j_N, j_M])(\varphi_{N,M}(d_N, d_M)).$$

The identity on A is $L(A) \xrightarrow{\text{id}} L(A) \xleftarrow{\text{id}} L(A)$, decorated by

$$d_A^{\text{id}} = \mathcal{D}(0 \rightarrow L(A))(\varphi_0(*)).$$

Theorem 12.3 (Interface category). *If $\mathbf{C}$ has chosen finite colimits, L is a left adjoint and strong symmetric monoidal for finite coproducts, and $\mathcal{D}_{r,\mathbb{T}}$ is the decoration datum of definition 12.1, then the objects of $\mathbf{I}_{\mathcal{T}}$ and morphisms of definition 12.2 form a category, up to the standard isomorphism classes. Disjoint union supplies a symmetric monoidal product when the decorating semantics is monoidal.*

Proof. Pushout associativity and the universal property give associativity up to canonical isomorphism; passing to isomorphism classes makes composition associative. Naturality and the associativity and unit axioms of the laxator give the same laws for d_P . Identity cospans and φ_0 give the unit laws. $\square$

This is the needed categorical construction, not decorative vocabulary. It is a BSC hybrid: structured cospans carry behavior and certificate decorations. Decorated and structured cospans are distinct constructions, with precise comparison hypotheses [Fong, 2015, Baez and Courser, 2020, Baez et al., 2022]. Heterogeneous internal types remain in the apex decoration; only declared ports are identified. Stochastic behavior may also be organized in a Markov category, where copying and discarding are structural but disintegration is not automatically available [Fritz, 2020]. Accordingly, equality-gluing of stochastic ports is legal here only when the chosen $\mathcal{D}_{\text{stoch},\mathbb{T}}$ explicitly supplies that composition; no conditional distribution or disintegration is inferred from the cospan alone.

12.2 Non-invertible transports and defects

Definition 12.4 (Fusion morphism). Let $(\mathcal{D}, \otimes, \mathbf{1}, \alpha, \lambda, \rho)$ be a k -linear additive monoidal category with finite biproducts. A finite fusion record is a finite family of defect objects $D_a \in \mathcal{D}$, including $D_1 \simeq \mathbf{1}$, nonnegative integers N_{ab}^c of finite support, and specified isomorphisms

$$D_a \otimes D_b \simeq \bigoplus_c D_c^{\oplus N_{ab}^c},$$

coherent with the associator and unit constraints; equivalently their two triple-fusion composites agree under the pentagon. If the defects act on a physical system A , the record additionally supplies a monoidal functor $\mathcal{D} \rightarrow \text{End}(A)$ in the declared ambient bicategory. Vector-space or lower-dimensional-TFT multiplicities require an explicitly tensored higher category and are a separate variant, not an untyped replacement for the integers above.

A quotient morphism pulls observables back only to the quotient-invariant subalgebra. A stochastic channel is generally irreversible and is compared by deficiency. A correspondence needs selection semantics. A fusion morphism branches, and the absence of an inverse is structural, not a failed group axiom. Physical use requires construction of the QFT, defect category, fusion, anomaly, and boundary conditions; schematic fusion does not validate a duality [Gaiotto et al., 2015, Choi et al., 2022, Apruzzi et al., 2023]. Symmetry-TFT constructions further organize non-invertible defect data in specified field theories; they do not supply a generic duality certificate [Kaidi et al., 2023].

13 Topological quantities and physical charge

Definition 13.1 (Topological charge). Let A be an abelian coefficient group. For $[\omega] \in H^k(M; A)$ and $[C] \in H_k(M; \mathbb{Z})$, the Kronecker evaluation is

$$Q_{\text{top}} = \langle [\omega], [C] \rangle \in A.$$

It is invariant under the stated homology/cohomology equivalences. In the differential-form case, $A = \mathbb{R}$, ω is a closed k -form, and the de Rham pairing is $Q_{\text{top}} = \int_C \omega$, with any normalization stated separately.

Definition 13.2 (Physical charge bridge). A physical charge is

$$q_{\text{phys}} = \chi(Q_{\text{top}}),$$

where $\chi : A \rightarrow Q_{\text{phys}}$ is a typed map into a declared physical quantity space with charge units, derived from at least one declared physical structure: gauge representation and coupling, an action and Noether current, a Gauss-law boundary flux, anomaly constraint, boundary condition, or calibrated measurement protocol. Its normalization and units are part of the derivation.

Proposition 13.3 (Normalization underdetermination). *Let the only supplied carrier data be a winding invariant $Q_{\text{top}} \in \mathbb{Z}$. These data do not identify a real-valued physical-charge map: they are compatible with every normalization $\chi_c : \mathbb{Z} \rightarrow \mathbb{R}$, $\chi_c(n) = cn$, $c \in \mathbb{R}$.*

Proof. For every real constant c , $\chi_c(n) = cn$ is a group homomorphism. The supplied data select none of the continuum of normalizations, nor the electric unit, sign convention, matter representation, or coupling to an electromagnetic current. A nonlinear or quotient-valued bridge introduces still more choices. $\square$

Winding can be a valid topological carrier while the physical promotion is blocked. Modern monopole–center-vortex work illustrates the full price of a bridge: it specifies $SU(N)$ gauge theory, compactification, 't Hooft flux, fractional topological charge, semiclassical action and fugacity, vacuum branches, anomaly constraints, and representation-sensitive string tensions in a controlled regime [Tanizaki and Ünsal, 2022, Hayashi and Tanizaki, 2024]. The adiabatic passage to strongly coupled undeformed $\mathbb{R}^4$ remains conjectural in that work; it does not provide hadron spectroscopy or validate an unrelated toroidal model.

14 Certificates, claim ledgers, and demotion

Definition 14.1 (Certificate schema). A certificate is a machine-readable and human-inspectable record

$$\text{Cert} = (\text{id}, \text{claim}, \text{types}, \text{assumptions}, \text{tolerances}, \text{sources}, \\ \text{artifacts}, \text{hashes}, \text{proof}, \text{execution}, \text{obligations}, \text{Stat}).$$

The proof field names the proposition and proof object or says “mathematical proof in text.” The execution field records code, inputs, environment, command, output, tolerances, and receipt, or says “unexecuted.” Hashes identify artifacts; they do not establish truth.

Definition 14.2 (Mechanical verification). A claim is mechanically verified only when a declared trusted kernel accepts a proof object for the encoded proposition in a pinned environment. The certificate records the kernel and prover version, theorem identifier, definitions, imported axioms, dependency hashes, build command, and accepted output. Kernel acceptance proves the encoded statement relative to those foundations; it does not prove that the encoding represents the intended physical system.

Lean supplies a small-kernel architecture and explicit proof objects [de Moura and Ullrich, 2021]; large formalizations show the value of modular dependency control [Hales et al., 2017, Commelin and Topaz, 2024]. A blueprint or proof DAG is not itself a proof.

Definition 14.3 (Numerical and empirical status). A numerical claim requires actual execution with code, inputs, environment, output, convergence and tolerance criteria, and hashes. A model description without execution is computationally unexecuted. An empirical claim also requires a sampling frame, instrument and calibration, protocol, exclusions, uncertainty analysis, and data provenance. Simulation agreement is not structural equivalence.

14.1 Corpus inheritance and local demotion

The complete supplied corpus was treated as evidence under [convention 1.3](#). Table 3 records the disposition of material not already represented by the retained tuples and morphism lineage. The words “retained” and “demoted” apply locally to the listed statement, not to an entire volume.

Table 3: Inheritance from the supplied BSC corpus.

Source	Retained mathematical content	Status boundary
Volume IV [Open Research Program, 2026a]	Identical observation laws imply a no-estimator obstruction; Fisher information additivity and data processing, rollout/cascade bounds, contraction criteria, entropy book-keeping, and $I_{\text{classical}} \leq I_{\text{quantum}}$ remain scoped mathematical templates. A boundary Poynting checksum must use an absolute residual or an interval enclosure.	The proposed optical and computational predictions are unexecuted here. Signed cancellation is not accepted as a conservation certificate.
Volumes V–VI [Open Research Program, 2026b,c]	Exact quotient descent, non-descent of noninvariant queries, and Blackwell comparison are retained where their hypotheses match this paper.	Composite human-grounding scores, scaffold claims, and empirical programs are unexecuted and untested; humane intent is not a mathematical bridge.
Second Catalogue and Volume VII [Tree, 2026a, Open Boundary-State Research Program, 2026]	Explicit repair discipline, bounded claims, and dependency-aware research questions inform the certificate grammar.	Civic, historical, and long-future meditations are outside the mathematical dependency graph and receive no physical promotion.
Arithmetic Recursive Folds [Tree, 2026b]	Finite sieve identities and the declared Mellin-transform backbone survive as ordinary arithmetic analysis.	Primitive-orbit language is an analogy unless an operator and trace formula are constructed; no Riemann-hypothesis proof is inherited.
Certified Arithmetic Trace Reconstruction [Tree, 2026c]	The finite-matrix prime-comb obstruction and the locally finite arithmetic-density construction are retained in their stated classes.	The prime-local operator fold is conjectural; finite truncations do not certify an infinite spectral correspondence.

Source	Retained mathematical content	Status boundary
Derived Witnessed Descent [J. Tree research program, 2026]	Strict, derived, and observed-derived holonomy are separate gates; a locally finite 0-chain has a natural atomic Radon-measure realization; the stated distributionally closed smooth counterterm class collapses.	No derived or observed gate is promoted by satisfying only the strict gate.
Formal Verification supplement [Unattributed, 2026]	The bounded-jet orthogonal-prime-block obstruction is retained only under its explicit hypotheses and quantifiers.	Mechanical runs described in the source are source-asserted, not replayed: the generating scripts and accepted JSON/proof receipts were not supplied. Their computational status is therefore unexecuted here.

14.2 Explicit demotion rules

1. An undefined domain, codomain, unit, quotient, or boundary condition demotes the mathematical axis to ill-posed.
2. A counterexample demotes the quantified mathematical claim to refuted; a repaired restricted claim receives a new identifier.
3. A proof sketch with an open lemma is conditional or conjectural, not proved.
4. Failure of a leakage gate invalidates promotion from the affected evaluation.
5. A numerical result without a reproducible receipt is unexecuted; a failed build is failed.
6. A verified preprint is not recorded as peer reviewed.
7. If an upstream dependency is demoted, descendants inherit the axiswise meet under definition 2.2.
8. Exceeding any prefix persistence tolerance blocks persistence at that horizon; later recovery does not erase the excursion.
9. A successful transfer outside its declared domain is local only; using it in a new domain requires a new morphism certificate.
10. An analogy may motivate a test but cannot occupy a mechanism dependency slot.

15 Proven formal consequences

This section collects consequences established in the present paper. Their scope is exactly the hypotheses cited; no empirical coordinate is promoted.

Theorem 15.1 (Admissible-chain bound). *For a compatible chain of stochastic BSC morphisms $\mathfrak{M}_{0,1}, \dots, \mathfrak{M}_{n-1,n}$, let η_i be the observation-channel contraction of the i -th post-processor and $\theta_i = \Theta_{i,i+1}$. Then*

$$\Theta_{0n} \leq \theta_{n-1} + \sum_{i=0}^{n-2} \theta_i \prod_{j=i+1}^{n-1} \eta_j.$$

For a common parameter family,

$$\delta(E_0, E_n) \leq \sum_{i=0}^{n-1} \delta(E_i, E_{i+1}).$$

Proof. Iterate theorems 7.5 and 7.6. □

Corollary 15.2 (Persistence obstruction). *If any prefix in an admissible fold chain exceeds its declared identity, viability, naturality, residual, or deficiency tolerance, that chain does not certify persistent identity through the prefix.*

Proof. This is the conjunction of definitions 7.2 and 9.2; the distinct coordinates are not allowed to compensate one another. $\square$

Theorem 15.3 (Operational factorization criterion). *Let $q : X \rightarrow \bar{X}$ be an exact operational quotient. A channel $\Phi : X \rightsquigarrow Z$ induces a unique channel $\bar{\Phi} : \bar{X} \rightsquigarrow Z$ if and only if Φ is constant on the fibers of q , under the standard Borel hypotheses of theorem 6.2.*

Proof. The forward implication follows from $\Phi = \bar{\Phi} \odot \delta_q$; the reverse implication is theorem 6.2. $\square$

Corollary 15.4 (Discard obstruction). *If two joint states have the same declared reduced quotient but different transformed accessible probabilities, no transform channel descends to that reduced quotient.*

Proof. Contrapositive of theorem 15.3. $\square$

Theorem 15.5 (DAG demotion monotonicity). *Repeated application of the axiswise meet rule on a finite claim DAG terminates at a unique greatest status assignment below the initial statuses and all dependency constraints. Status can only remain fixed or demote unless new independent evidence is added.*

Proof. Each finite status axis is a finite meet-semilattice. Processing vertices in topological order applies each predecessor constraint after its inputs have stabilized. The coordinatewise meet is associative, commutative, and idempotent, yielding the unique greatest feasible assignment. $\square$

Corollary 15.6 (Topology-to-charge block). *A computed topological class does not promote a physical-charge claim unless a certified bridge χ is present as a dependency.*

Proof. Apply proposition 13.3 and theorem 15.5. $\square$

16 Worked fixtures and counterexamples

Each fixture is intentionally small. It fixes inputs and expected status so that future implementations can be tested against a stable result.

Worked fixture 16.1 (Torus winding without electric charge). **Inputs and types.** Let $T^2 = S^1 \times S^1$ and

$$\gamma : [0, 2\pi] \rightarrow T^2, \quad \gamma(t) = (e^{2it}, e^{-3it}).$$

The admissible observable is the pair of winding integrals

$$w_j(\gamma) = \frac{1}{2\pi i} \int_{\gamma_j} z^{-1} dz \in \mathbb{Z}.$$

Assumptions. Endpoints are identified, orientation is increasing t , and homology has coefficients $\mathbb{Z}$. **Claimed result.**

$$[\gamma] = (2, -3) \in H_1(T^2; \mathbb{Z}) \cong \mathbb{Z}^2.$$

Test. Substitute $z_1 = e^{2it}$, $z_2 = e^{-3it}$ in the integrals, or reduce the lifted endpoint increments by 2π . **Expected certificate.** Mathematical status proved; source present calculation; computation not required; transfer to electric charge blocked. **Failure and demotion.** Any output q_{electric}

without a typed $\chi : \mathbb{Z}^2 \rightarrow \mathbb{R}C$ fails [definition 13.2](#). **Evidence that changes the result.** A gauge/action/current or calibrated measurement derivation of χ , with normalization, units, matter representation, and anomaly checks, can open the physical-charge dependency; it cannot change the homology result.

Worked fixture 16.2 (One-dimensional Dirichlet-to-Neumann ambiguity). **Inputs and types.**

Let $\gamma \in L^\infty(0,1)$ satisfy $0 < \gamma_- \leq \gamma \leq \gamma_+ < \infty$, and solve

$$-(\gamma u')' = 0, \quad u(0) = f_0, \quad u(1) = f_1.$$

The trace is $\tau_\partial u = (u(0), u(1)) \in \mathbb{R}^2$; extension sends $f \in \mathbb{R}^2$ to $u_f \in H^1(0,1)$; the response is $\Lambda_\gamma : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and sends f to outward flux. Put $R_\gamma = \int_0^1 \gamma(x)^{-1} dx$. **Claimed result.**

$$\Lambda_\gamma = \frac{1}{R_\gamma} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

The distinct conductivities

$$\gamma_1 \equiv 1, \quad \gamma_2(x) = \begin{cases} 2, & 0 < x < 1/2, \\ 2/3, & 1/2 < x < 1 \end{cases}$$

both have $R_\gamma = 1$, hence identical response. **Test.** Constant flux gives $\gamma u' = (f_1 - f_0)/R_\gamma$; apply outward signs at the endpoints. **Noise bound.** In Euclidean operator norm,

$$\|\Lambda_R - \Lambda_{\tilde{R}}\|_2 = 2|R^{-1} - \tilde{R}^{-1}|.$$

For $v = (1, -1)^\top$, define the estimator

$$\hat{g} = \frac{1}{4} v^\top \hat{\Lambda} v.$$

Then $\|\hat{\Lambda} - \Lambda_R\|_2 \leq \eta$ implies

$$|\hat{g} - R^{-1}| \leq \frac{1}{4} \|v\|_2^2 \eta = \eta/2.$$

Thus the data identify only effective conductance $g = R^{-1}$ to this error, not the spatial profile. **Expected certificate.** Forward solution and DN formula proved; profile reconstruction non-identifiable; noise statement bounded. **Failure and demotion.** “The boundary reconstructs $\gamma(x)$ ” is refuted for this class. **Evidence that changes the result.** A restricted parametric class, interior data, additional frequencies, or a proved injective data map with stability can yield a new scoped claim.

Worked fixture 16.3 (Exact Blackwell order and directed deficiency). **Inputs and types.** Let $\Theta = \{0,1\}$ and $Y_\ell = Y_m = \{0,1\}$. The source experiment is perfect:

$$P_{\ell,0} = (1,0), \quad P_{\ell,1} = (0,1).$$

The target is a binary symmetric garbling:

$$P_{m,0} = (3/4, 1/4), \quad P_{m,1} = (1/4, 3/4).$$

Assumptions. Total variation is half the ℓ^1 distance; all Markov matrices are allowed. **Claimed result.**

$$\delta(E_\ell, E_m) = 0, \quad \delta(E_m, E_\ell) = 1/4, \quad \Delta(E_\ell, E_m) = 1/4.$$

Test. The 1/4-crossover channel maps E_ℓ exactly to E_m . Conversely let r_y be the probability of output 1 after observing y under a proposed reverse channel. Its two probabilities are

$p_0 = (3r_0 + r_1)/4$ and $p_1 = (r_0 + 3r_1)/4$, so $p_1 - p_0 = (r_1 - r_0)/2 \leq 1/2$. Errors below $1/4$ would require $p_0 < 1/4$ and $p_1 > 3/4$, a contradiction. The identity reverse channel attains $1/4$. **Expected certificate.** Exact rational proof; mathematical status proved; source and target direction explicit. **Failure and demotion.** Reversing the prose or calling directed deficiency symmetric is a type/status failure. **Evidence that changes the result.** Enlarging the observation or restricting the parameter/decision class defines a new experiment and may change the deficiency.

Worked fixture 16.4 (Koopman pseudomode and spectral pollution). **Inputs and types.** Let $X = \{-1, 1\}^{\mathbb{Z}}$ with product measure and the two-sided shift $F(x)_j = x_{j+1}$. Put $\phi_j(x) = x_j$, so $\mathcal{K}_F \phi_j = \phi_{j+1}$ and the ϕ_j are orthonormal in $L^2(\mu)$. For $|\lambda| = 1$,

$$\psi_N = N^{-1/2} \sum_{j=0}^{N-1} \lambda^{-j} \phi_j.$$

Claimed result.

$$\|\mathcal{K}_F \psi_N - \lambda \psi_N\|_2 = \sqrt{2/N}.$$

Yet compression to $\mathcal{G}_N = \text{span}\{\phi_0, \dots, \phi_{N-1}\}$ is a nilpotent unilateral truncation with every eigenvalue 0, whereas the true Koopman operator is unitary and $0 \notin \sigma(\mathcal{K}_F)$. **Test.** Interior coefficients telescope; only orthogonal endpoint terms $-\lambda \phi_0 / \sqrt{N}$ and $\lambda^{-(N-1)} \phi_N / \sqrt{N}$ remain. Direct matrix compression gives a nilpotent shift. **Expected certificate.** Symbolic residual proved; the zero eigenvalues are certified spectral pollution; no empirical execution claimed. **Failure and demotion.** Interpreting ψ_N as an exact eigenmode, or compression eigenvalue 0 as physical decay, is refuted. **Evidence that changes the result.** A residual-controlled limit in a declared topology, perturbation stability, and an applicable Koopman learnability theorem can promote a specified spectral claim.

Worked fixture 16.5 (Finite $\mathbb{Z}_2$ quantum-reference-frame descent). **Inputs and types.** Let $G = \mathbb{Z}_2$, $\mathcal{H}_R = \ell^2(G)$, and $\mathcal{H}_S = \mathbb{C}^2$. The frame carries the left-regular action $L_h|g\rangle = |h+g\rangle$, and the system carries $U_S(h) = X^h$. Define the frame-trivialization unitary

$$W = \sum_{g \in G} |g\rangle\langle g|_R \otimes X_S^g.$$

Write $\text{Ad}_W(\rho) = W\rho W^\dagger$. The target accessible algebra is $I_R \otimes \mathcal{B}(\mathcal{H}_S)$. Its paired pullback is the relational algebra

$$W^\dagger(I_R \otimes A)W = \sum_{g \in G} |g\rangle\langle g|_R \otimes X^g A X^g,$$

which is invariant under the simultaneous action $L_h \otimes X^h$. Let

$$\rho_1 = | +0\rangle\langle +0|, \quad \rho_2 = |00\rangle\langle 00|.$$

Both have reduced system state $|0\rangle\langle 0|$. **Claimed result.**

$$\text{Tr}_R(W\rho_1 W^\dagger) = I_S/2, \quad \text{Tr}_R(W\rho_2 W^\dagger) = |0\rangle\langle 0|.$$

Their trace distance is $1/2$; the expectations of Z_S are 0 and 1. Hence no channel on the reduced system can factor $\text{Tr}_R \circ \text{Ad}_W$ through Tr_R on a class containing ρ_1, ρ_2 . **Test.** Here W is $\text{CNOT}_{R \rightarrow S}$. It maps $| +0\rangle$ to the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$. Discarding R gives $I/2$. It leaves $|00\rangle$ unchanged. Equivalently, for the discard–reset–transform route with $\iota_0(\rho_S) = |0\rangle\langle 0|_R \otimes \rho_S$,

$$\text{Tr}_R \text{Ad}_W(\iota_0(\text{Tr}_R \rho_1)) = |0\rangle\langle 0| \neq I/2.$$

Expected certificate. Exact 4×4 matrix or ket calculation; group action, paired relational algebra, and nonfactorization proved on the declared accessible algebra. **Failure and demotion.** No reduced transformed channel may be inferred from $\text{Tr}_R \rho$ alone on a class containing joint correlations that alter the output. **Evidence that changes the result.** A restricted state class, final-frame localizability condition, or proved factorization through the quotient can make the operations agree. **Scope.** This is a finite-group QRF fixture, not a universal model of all quantum reference frames.

Worked fixture 16.6 (Locally satisfiable parity data that do not glue). **Inputs and types.** Let vertex variables $x_i \in \mathbb{Z}_2$ on a 3-cycle and relation stalks

$$R_{12} : x_1 + x_2 = 0, \quad R_{23} : x_2 + x_3 = 0, \quad R_{31} : x_3 + x_1 = 1.$$

Restriction from an edge relation to either endpoint is coordinate projection. **Assumptions.** Addition is modulo 2. **Claimed result.** Every edge relation is nonempty and every endpoint projection is $\{0, 1\}$, but the inverse limit of the three relations is empty. **Test.** Adding all equations gives $2(x_1 + x_2 + x_3) = 1$ in $\mathbb{Z}_2$, hence $0 = 1$. Equivalently the parity cocycle $(0, 0, 1)$ has cycle sum 1 and is the nonzero class in $H^1(C_3; \mathbb{Z}_2)$. **Expected certificate.** Exact finite contradiction and a declared coefficient complex; local validity preserved, global-section claim refuted. **Failure and demotion.** Individual plausibility or surjective overlap projections do not promote a global state. **Evidence that changes the result.** Altering one parity, adding a slack/noise model with a stated consistency radius, or identifying a faulty edge yields a different scoped gluing problem.

Worked fixture 16.7 (Off-shell massive-field shift). **Inputs and types.** All variables in this fixture are nondimensionalized. On $\Omega = (0, 1)$, put

$$H_N^2(0, 1) = \{\phi \in H^2(0, 1) : \phi'(0) = \phi'(1) = 0\}$$

and define

$$\mathcal{E}_m : H_N^2(0, 1) \rightarrow L^2(0, 1), \quad \mathcal{E}_m(\phi) = -\phi'' + m^2\phi, \quad m > 0.$$

The source equation is $\mathcal{E}_m(\phi) = 0$. Let $T_c\phi = \phi + c$, $c \in \mathbb{R}$. **Claimed result.**

$$R_c(\phi) = \mathcal{E}_m(T_c\phi) - \mathcal{E}_m(\phi) = m^2c, \quad \|R_c\|_{L^2(0,1)} = m^2|c|.$$

Neumann boundary conditions are preserved, but the target field equation is not for $c \neq 0$. **Test.** Differentiate the constant and substitute. **Expected certificate.** Exact dimensionless residual, source equation and target equation stated; alleged symmetry blocked off shell. **Failure and demotion.** Geometric resemblance or boundary compatibility cannot compensate for nonzero equation residual. **Evidence that changes the result.** The cases $c = 0$, $m = 0$, or a modified action/source transformation define different claims; an on-shell map requires a proof for the full solution class.

Worked fixture 16.8 (Permanent exact regression counterexample). **Inputs and types.** The quantified claim is $\forall x \in \mathbb{R}, \sqrt{x^2} = x$. The exact integer input is $x = -1$; the principal square root is computed as $\text{isqrt}(x^2) = 1$. **Claimed result.** $1 \neq -1$, so the quantified claim is false; the correct real identity is $\sqrt{x^2} = |x|$. **Mechanical test.** The retained script `verify_counterexample.py`, run under CPython 3.12.13, emitted a deterministically serialized, key-sorted JSON receipt twice with byte-identical output. The script SHA-256 is `0c5e019d3398b4aacfeeee29ef05e97664989b3412459eefab05d01c1bc84ada`; the receipt SHA-256 is `7b7d416914068b1e1e5510b5658f82d17c9615bfa6954ec5facfb6eeb0f43333`. **Expected certificate.** Mathematical status refuted for the universal claim; computational status exact receipt; no floating arithmetic. **Failure and demotion.** Any future rewrite that returns the original universal identity passes formatting but fails the permanent regression. **Evidence that changes the result.** Restricting $x \geq 0$ creates a true new proposition; it does not repair the original quantified claim.

17 Application boundaries: holography, QCD, primordial-field theory, and learned models

17.1 Holography and reconstruction

BSC can audit a holographic claim, but does not prove holography. A valid application must identify the bulk and boundary theories, state spaces and observable algebras, code subspace, reconstruction map, gauge quotient, semiclassical or large-parameter limit, error topology, and physical dictionary. An inverse-PDE uniqueness theorem and a holographic duality are different claims. The former cannot be promoted by the visual fact that data reside on a boundary; the latter cannot be promoted by the generic quotient theorems in this paper.

17.2 QCD and controlled confinement comparisons

The monopole–center-vortex constructions cited in [section 13](#) provide a stronger comparator because their carrier and bridge are internal to a specified gauge theory. Even there, controlled compactified regimes, anomaly arguments, condensation criteria, and string tensions do not automatically supply physical $\mathbb{R}^4$ QCD, generations, electroweak charge, hadron spectra, or scattering. A BSC certificate must preserve that domain boundary.

17.3 Klingman’s primordial-field papers

The primordial-field papers propose a candidate structure consisting of a self-interacting field, dual field components, toroidal configurations, complex or Kähler duality, closed paths and winding, stabilization through competing effects, a winding-to-charge correspondence, and a solenoidal confinement mechanism [[Klingman, 2024, 2022](#)]. These are legitimate objects for formalization and testing. The current classification is:

Candidate grammar: topological carrier and stabilization model.

Not established: a derivation of quark charge, QCD, or hadron dynamics.

The papers’ toroidal parametrization supplies a winding computation, but a parametric flow is not yet a solution generated by a well-typed physical action. The exact bridge still owed is displayed in [table 4](#).

Failure of promotion does not erase useful mathematics: winding classes, stability conditions, and toroidal boundary-value problems can survive as mathematical candidate components. BSC does not validate primordial-field theory, and the theory does not validate BSC.

17.4 Learned models

A neural operator, dynamic mode model, or surrogate is a fitted transport with a restricted dictionary and data law. Its certificate must separate training fit, held-out prediction, full-equation residual, conservation closure, distribution shift, scale transfer, and implementation. Spectral pollution, non-identifiability, and adversarial learnability limits remain live even when visual predictions are persuasive. Conversely, an impossibility theorem for a broad task class does not demote a model with stronger verified structure.

17.5 Large proof programs as scope controls

Two contemporary cases illustrate why source status and quantifiers matter. The Wang–Zahl proof of the Keakeya set conjecture in $\mathbb{R}^3$, with a later streamlined proof, remains a preprint pro-

Table 4: Bridge obligations for physical promotion of the primordial-field candidate.

Required object	Inspectable test	Current status
Action and field space	variational derivative; dimensions; boundedness or domain of validity	not supplied in a form deriving the claimed torus
Gauge covariance and conserved current	transformation law, Noether/Gauss identity, charge normalization	bridge χ absent
Boundary and regularity class	well-posed field equations and finite energy	not established for the full candidate
Anomaly cancellation or matching	fermion content and anomaly polynomial or controlled matching condition	not established
Spin and statistics	representation of the Lorentz/rotation group and quantization rule	not established
Color structure	$SU(3)$ representations, gauge connection, observables	not established
Confinement criterion	Wilson/'t Hooft behavior or gauge-invariant equivalent with regime	solenoidal analogy only
Running and scale	renormalization prescription and beta function	not established
Generations and mixing	spectrum and mixing matrix from the action	not established
Spectra and scattering	hadron masses, amplitudes, uncertainties, and independent falsifiers	not established

gram at the checked date and does not settle dimensions $n \geq 4$ or all adjacent harmonic-analysis conjectures [Wang and Zahl, 2025, Guth et al., 2026]. The five-paper geometric Langlands program states a proof of the categorical global unramified conjecture in its characteristic-zero setting; the checked core papers are preprints and do not prove the arithmetic Langlands program or every local and ramified form [Gaitsgory and Raskin, 2024a, Arinkin et al., 2024a, Campbell et al., 2024, Arinkin et al., 2024b, Gaitsgory and Raskin, 2024b]. These examples are not evidence for BSC. They are models of exact scope, source labeling, and large dependency architecture.

18 Falsifier matrix

Table 5: Predeclared falsifiers and local demotion behavior.

Claim class	Decisive adverse finding	Immediate demotion	What remains
Boundary sufficiency	positive exterior residual above threshold after legal conditioning	transfer blocked for that target/domain	boundary may remain predictive but insufficient
Interior reconstruction	two inequivalent admissible interiors with identical data	identifiability refuted	forward response and quotient recovery may remain valid
Operational quotient	query differs on one exact observation class	query descent refuted	quotient valid for other descending queries
Scale transfer	target residual or prefix defect exceeds tolerance	transfer blocked at that scale/horizon	source and target models remain local
Persistent identity	any required prefix leaves identity or viability tube	persistence through that horizon refuted	endpoint recurrence may still hold
Koopman mode	certified full residual lower bound exceeds tolerance, or polluted eigenvalue rejected	mode claim demoted	finite predictor may still have empirical utility
Frame-change discard	two quotient-equal inputs give different transformed accessible probabilities	reduced map nonexistent on that class	paired full-system transformation remains valid
Local-to-global state	empty inverse limit or nonzero declared obstruction	global claim refuted	each local section may remain valid
Topological electric charge	two physically admissible normalizations or no derived χ	charge promotion blocked	topological invariant remains
Primordial-field QCD	failure of gauge covariance, anomaly consistency, spectra, or an independent prediction	affected physical bridge refuted	toroidal mathematics may remain
Numerical verification	irreproducible build, hash mismatch, or tolerance failure	computational axis failed	mathematical statement unaffected unless execution was its only support
Mechanical proof	kernel rejection or untracked axiom/dependency	mechanical status withdrawn	informal proof may be assessed separately

19 Open problems and highest-leverage next proofs

Open problem 19.1 (Observable-complete morphism category). Construct a category or double category whose objects are $(\hat{S}_\ell, \mathcal{C}_\ell)$, whose horizontal arrows are physical interfaces, whose vertical arrows are one fixed morphism variant, and whose 2-cells certify paired observable transport. Prove closure of hard gates and identity normalization without assuming the needed coherence.

Open problem 19.2 (Residual calculus for stochastic equations). For nonlinear stochastic

PDEs, define a weak equation-space transport S_{ℓ_m} and residual that is stable under law-level composition. Prove an analogue of [theorem 7.7](#) without replacing nonlinear residuals by an unjustified expectation.

Open problem 19.3 (Persistence under perturbation). Find checkable conditions under which prefix operational equivalence, viability, and deficiency remain uniformly bounded over long horizons. Separate invariant laws, periodic laws, quasi-stationarity, metastability, and return recurrence.

Open problem 19.4 (Boundary reconstruction with instrument uncertainty). Combine PDE stability with uncertainty in the observation kernel, finite boundary layer, calibration, and geometry. The goal is an identified set and stability modulus that accounts for all four sources without treating them as one noise scalar.

Open problem 19.5 (Operational QRF descent). Characterize the state classes and accessible algebras for which a quantum reference-frame transformation factors through a declared subsystem quotient. Relate final-frame localizability to the factorization criterion of [theorem 15.3](#).

Open problem 19.6 (Learnability-aware recurrence). Join residual Koopman certification, perturbation pseudospectra, and SCI classification to produce a recurrence certificate with an honest ordered limit and finite-data stopping rule.

Open problem 19.7 (Physical topology-to-charge bridges). For any proposed carrier, derive χ from a complete action/gauge/current or measurement structure and test anomaly, boundary, quantization, and normalization conditions. A no-go result is as valuable as a construction.

Open problem 19.8 (Machine-checked BSC kernel). Formalize the status lattice, dependency demotion, finite experiment deficiency fixture, parity-gluing obstruction, and morphism chain inequalities in a declared proof assistant. Pin the environment and retain accepted proof objects and build receipts.

The highest-leverage order is: quotient and observable descent; morphism composition with equation-space residuals; prefix persistence; instrumented boundary stability; then domain-specific physical bridges. This order attacks hard obstructions before expensive deployment.

20 Conclusion

BSC earns its place by refusing silent transfers. Its retained system object describes a finite, open, instrumented domain. Its measurable repair separates raw states from equivalence, supplies context and legal filtration, and exposes the dynamics and observation kernels. The added morphism record makes state transport travel with observables, records the information destroyed by post-processing and quotient, tests target equations and commuting diagrams, requires a physical completion, and carries its own certificate.

The proved content is local but useful: quotient-respecting kernels descend; paired state and observable transport preserves accessible probabilities; statistical deficiency and naturality defects obey composition bounds; equation residuals propagate when an equation-space map exists; compatible sheaf sections glue while locally satisfiable relations may fail globally; dependency demotion is monotone; and topology alone does not select electric charge. The fixtures make these statements vulnerable to exact tests.

Persistence is consequently neither visual recurrence nor final-time return. It is maintenance of a declared operational identity through every required fold, within viability, residual, loss, and evidence budgets. This principle may organize future theories; it has not been promoted to a universal law.

A representation may be beautiful. A mechanism must be typed. A transfer must account for its loss. A recurrence must survive perturbation. A computation must leave a receipt. A physical claim must pay the full price of its bridge.

References

- Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011. doi: 10.1088/1367-2630/13/11/113036.
- Shadi Ali Ahmad, Thomas D. Galley, Philipp A. Höhn, Maximilian P. E. Lock, and Alexander R. H. Smith. Quantum relativity of subsystems. *Physical Review Letters*, 128:170401, 2022. doi: 10.1103/PhysRevLett.128.170401.
- Giovanni Alessandrini. Stable determination of conductivity by boundary measurements. *Applicable Analysis*, 27:153–172, 1988. doi: 10.1080/00036818808839730.
- Fabio Apruzzi, Federico Bonetti, Iñaki García Etxebarria, Saghar S. Hosseini, and Sakura Schäfer-Nameki. Symmetry TFTs from string theory. *Communications in Mathematical Physics*, 402(1):895–949, 2023. doi: 10.1007/s00220-023-04737-2.
- D. Arinkin, D. Beraldo, J. Campbell, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin, and N. Rozenblyum. Proof of the geometric Langlands conjecture ii: Kac–Moody localization and the FLE, 2024a. Preprint.
- D. Arinkin, D. Beraldo, L. Chen, J. Faergeman, D. Gaitsgory, K. Lin, S. Raskin, and N. Rozenblyum. Proof of the geometric Langlands conjecture iv: Ambidexterity, 2024b. Preprint.
- Scott N. Armstrong and Charles K. Smart. Quantitative stochastic homogenization of convex integral functionals. *Annales Scientifiques de l’École Normale Supérieure*, 49(2):423–481, 2016. doi: 10.24033/asens.2287.
- Kari Astala and Lassi Päiväranta. Calderón’s inverse conductivity problem in the plane. *Annals of Mathematics*, 163:265–299, 2006. doi: 10.4007/annals.2006.163.265.
- John C. Baez and Kenny Courser. Structured cospans. *Theory and Applications of Categories*, 35: 1771–1822, 2020.
- John C. Baez, Kenny Courser, and Christina Vasilakopoulou. Structured versus decorated cospans. *Compositionality*, 4:3, 2022. doi: 10.32408/compositionality-4-3.
- Claude Bardos, François Golse, and C. David Levermore. Fluid dynamic limits of kinetic equations. ii. convergence proofs for the Boltzmann equation. *Communications on Pure and Applied Mathematics*, 46:667–753, 1993. doi: 10.1002/cpa.3160460503.
- Alain Bensoussan, Jacques-Louis Lions, and George C. Papanicolaou. *Asymptotic Analysis for Periodic Structures*. North-Holland, Amsterdam, 1978.
- David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032.
- Boundary-State Calculus. Mathematical completion modules for trans-scale boundary-state field calculus, July 2026a. Supplied internal manuscript; source status: internal.
- Boundary-State Calculus. Trans-scale boundary-state field calculus, volume iii: Forward mathematics, testable consequences, and completion of the audit program, July 2026b. Supplied internal manuscript; source status: internal.

- Boundary-State Research Program. Trans-scale boundary-state field calculus: Typed kernels, boundary observation, scale transfer, quantum edge response, terminal cosmology, and biophysical boundary modules, July 2026. Supplied internal manuscript; source status: internal.
- Justin Campbell, Lin Chen, Joakim Faergeman, Dennis Gaitsgory, Kevin Lin, Sam Raskin, and Nick Rozenblyum. Proof of the geometric Langlands conjecture iii: Compatibility with parabolic induction, 2024. Preprint.
- Titouan Carrette, Jan Głowacki, and Leon Loveridge. Operational quantum reference frame transformations. *Quantum*, 9:1680, 2025. doi: 10.22331/q-2025-03-27-1680.
- Esteban Castro-Ruiz and Ognian Oreshkov. Relative subsystems and quantum reference frame transformations. *Communications Physics*, 8:187, 2025. doi: 10.1038/s42005-025-02036-x.
- Yichul Choi, Clay Córdova, Po-Shen Hsin, Ho Tat Lam, and Shu-Heng Shao. Noninvertible duality defects in $3 + 1$ dimensions. *Physical Review D*, 105:125016, 2022. doi: 10.1103/PhysRevD.105.125016.
- Matthew J. Colbrook and Anders C. Hansen. The foundations of spectral computations via the solvability complexity index hierarchy. *Journal of the European Mathematical Society*, 25: 4639–4718, 2023. doi: 10.4171/JEMS/1289.
- Matthew J. Colbrook and Alex Townsend. Rigorous data-driven computation of spectral properties of Koopman operators for dynamical systems. *Communications on Pure and Applied Mathematics*, 77(1):221–283, 2024. doi: 10.1002/cpa.22125.
- Matthew J. Colbrook, Lorna J. Ayton, and Máté Szőke. Residual dynamic mode decomposition: Robust and verified Koopmanism. *Journal of Fluid Mechanics*, 955:A21, 2023. doi: 10.1017/jfm.2022.1052.
- Matthew J. Colbrook, Igor Mezić, and Alexei Stepanenko. Adversarial dynamical systems characterize when data-driven learning succeeds or fails. *Nature Communications*, 17:5397, 2026. doi: 10.1038/s41467-026-74220-8. Preprint title: Limits and Powers of Koopman Learning, arXiv:2407.06312.
- Johan Commelin and Adam Topaz. Abstraction boundaries and spec driven development in pure mathematics. *Bulletin of the American Mathematical Society*, 61(2):241–255, 2024. doi: 10.1090/bull/1831.
- Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021. doi: 10.1007/978-3-030-79876-5_37.
- Yu Deng, Zaher Hani, and Xiao Ma. Long time derivation of the Boltzmann equation from hard sphere dynamics, 2024. Preprint.
- Yu Deng, Zaher Hani, and Xiao Ma. Hilbert’s sixth problem: Derivation of fluid equations via Boltzmann’s kinetic theory, 2025. Preprint.
- Brendan Fong. Decorated cospans. *Theory and Applications of Categories*, 30:1096–1120, 2015.
- Tobias Fritz. A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics. *Advances in Mathematics*, 370:107239, 2020. doi: 10.1016/j.aim.2020.107239.

- Davide Gaiotto, Anton Kapustin, Nathan Seiberg, and Brian Willett. Generalized global symmetries. *Journal of High Energy Physics*, 2015(2):172, 2015. doi: 10.1007/JHEP02(2015)172.
- Dennis Gaitsgory and Sam Raskin. Proof of the geometric Langlands conjecture i: Construction of the functor, 2024a. Preprint.
- Dennis Gaitsgory and Sam Raskin. Proof of the geometric Langlands conjecture v: The multiplicity one theorem, 2024b. Preprint.
- Antoine Gloria and Felix Otto. An optimal error estimate in stochastic homogenization of discrete elliptic equations. *The Annals of Applied Probability*, 22(1):1–28, 2012. doi: 10.1214/10-AAP745.
- François Golse and Laure Saint-Raymond. The Navier–Stokes limit of the Boltzmann equation for bounded collision kernels. *Inventiones Mathematicae*, 155:81–161, 2004. doi: 10.1007/s00222-003-0316-5.
- Larry Guth, Hong Wang, and Joshua Zahl. A Streamlined Proof of the Kakeya Set Conjecture in $\mathbb{R}^3$, 2026. Preprint.
- Thomas Hales, Mark Adams, Gertrud Bauer, Tat Dat Dang, John Harrison, Truong Le Hoang, Cezary Kaliszyk, Victor Magron, Sean McLaughlin, Tat Thang Nguyen, Truong Quang Nguyen, Tobias Nipkow, Steven Obua, Joseph Pleso, Jason Rute, Alexey Solovyev, An Hoai Thi Ta, Nam Trung Tran, Thi Diep Trieu, Josef Urban, Ky Vu, and Roland Zumkeller. A formal proof of the Kepler conjecture. *Forum of Mathematics, Pi*, 5:e2, 2017. doi: 10.1017/fmp.2017.1.
- Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3:315–358, 2019. doi: 10.1007/s41468-019-00038-7.
- Yui Hayashi and Yuya Tanizaki. Unifying monopole and center vortex as the semiclassical confinement mechanism. *Physical Review Letters*, 133:171902, 2024. doi: 10.1103/PhysRevLett.133.171902.
- J. Tree research program. Derived witnessed descent and atomic spectral complexity: A corrected mathematical bridge for boundary-state field calculus. Supplied internal Markdown research note dated 21 July 2026; source status: theorem package with explicit open problems; unpublished, July 2026.
- Justin Kaidi, Kantaro Ohmori, and Yunqin Zheng. Symmetry TFTs for non-invertible defects. *Communications in Mathematical Physics*, 404:1021–1124, 2023. doi: 10.1007/s00220-023-04859-7.
- Edwin Eugene Klingman. Higher order self-induced self-interacting field. *Journal of High Energy Physics, Gravitation and Cosmology*, 8(2):285–302, 2022. doi: 10.4236/jhepgc.2022.82023. Primary proposal; no independent physical validation inferred.
- Edwin Eugene Klingman. The origin of quarks in quantum gravity. *Journal of Modern Physics*, 15(8):1229–1245, 2024. doi: 10.4236/jmp.2024.158050.
- Bernard O. Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931. doi: 10.1073/pnas.17.5.315.
- Matti Lassas and Gunther Uhlmann. On determining a Riemannian manifold from the Dirichlet-to-Neumann map. *Annales Scientifiques de l’École Normale Supérieure*, 34(5):771–787, 2001. doi: 10.1016/S0012-9593(01)01076-X.

Lucien Le Cam. Sufficiency and approximate sufficiency. *The Annals of Mathematical Statistics*, 35(4):1419–1455, 1964. doi: 10.1214/aoms/1177700372.

Open Boundary-State Research Program. Trans-scale boundary-state field calculus, volume vii: The seventh boundary: Thirty-three meditations for the long human future, 2027–2059: A human covenant for intelligence, dignity, attention, work, creation, relationship, conscience, ecology, public abundance, and joy. Supplied internal research and civic synthesis draft; unpublished, July 2026.

Open Research Program. Trans-scale boundary-state field calculus, volume iv: Optical-computational field theory, scientific machine learning, and closed-loop experimental intelligence: Maxwell boundary objects, coherent and incoherent observation kernels, inverse problems, neural operators, photonic computing, active experiment design, and falsifiable optical-AI predictions. Supplied internal research thesis draft; unpublished, July 2026a.

Open Research Program. Trans-scale boundary-state field calculus, volume v: Living identity, durable intelligence, and scaffold ecology: Human grounding, cognitive offloading, agency, reality surplus, reflexive observation, institutional resilience, and falsifiable human-AI systems. Supplied internal research thesis draft; unpublished, July 2026b.

Open Research Program. Trans-scale boundary-state field calculus, volume vi: Reality surplus: A human-centered theory of grounded understanding: Quotient reach, structural non-exhaustion, decision descent, evidence refinement, contestable action, and falsifiable human-AI systems. Supplied internal research thesis draft; unpublished, July 2026c.

John Sylvester and Gunther Uhlmann. A global uniqueness theorem for an inverse boundary value problem. *Annals of Mathematics*, 125(1):153–169, 1987. doi: 10.2307/1971291.

Yuya Tanizaki and Mithat Ünsal. Center vortex and confinement in Yang–Mills theory and QCD with anomaly-preserving compactifications. *Progress of Theoretical and Experimental Physics*, 2022:04A108, 2022. doi: 10.1093/ptep/ptac042.

J. Tree. On the living future: A boundary-state synthesis of flourishing, repair, and civilizational continuity, 2027–3033, July 2026a. Supplied internal synthesis; source status: internal.

J. Tree. Arithmetic recursive folds: Prime generation, mellin resonance, and the critical scale boundary. Supplied independent research manuscript; unpublished, July 2026b.

J. Tree. Certified arithmetic trace reconstruction beyond finite matrices: Effective weil dictionaries, relative operator limits, and optical boundary-state benchmarks. Supplied independent research manuscript; unpublished, July 2026c.

Unattributed. Formal verification of derived holonomy, a mechanical shifted-ladder witness, and an orthogonal prime-block no-go. Supplied unattributed internal Markdown mathematical supplement dated 21 July 2026; unpublished, July 2026.

Hong Wang and Joshua Zahl. Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions, 2025. Preprint.